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Present-Day Climate and Climate Sensitivity in the Meteorological Research Institute Coupled GCM Version 2.3 (MRI-CGCM2.3)

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Abstract

A new version of the Meteorological Research Institute (MRI) coupled general circulation model MRI-CGCM2 (MRI2.3) is developed and compared with the previous version (MRI2.0). The cloud scheme includes diagnostic function for cloud amount separately specified for convective and layer clouds, which is one of the major modifications contributing to the improved model performance. MRI2.3 exhibits better agreement with the observations in many aspects of present-day climate simulations, including the global energy budget, meridional distributions of shortwave and longwave radiation at the top of the atmosphere, and geographical distributions of surface air temperature and precipitation. The effective climate sensitivity of each version is evaluated based on an experiment with a transient (1%/year) increase of carbon dioxide concentration. The effective climate sensitivity of MRI2.3 (2.9 K) is about twice that of MRI2.0 (1.4 K). The change in the cloud-forcing response, particularly for shortwave cloud forcing, is essential for increasing climate sensitivity. A difference in tropical low-level clouds over the subsidence regions contributes significantly to the difference in cloud-forcing changes in response to a climate change. Analyses based on circulation regimes, defined by the vertical velocity at the mid-troposphere, suggest that the cloud-forcing response in the tropics is controlled more by thermodynamic characteristics, such as changes of the stability in the lower troposphere, rather than by large-scale circulation changes, such as a change in the subsidence strength.

1. Introduction

The second-generation Coupled General Circulation Model (MRI-CGCM2) was developed at the Meteorological Research Institute (MRI), and released as version MRI-CGCM2.0 (Yukimoto et al. 2001). The model was used for climate-change simulations, and some of the results were included in the Third Assessment Report (TAR) of the Intergovermental Panel on Climate Change (IPCC 2001). The IPCC TAR reported that the inter-model scatter of climate sensitivity was still substantial, ranging between 2.0 K and 5.1 K, and there was almost no reduction from the previous range of 1.9 to 5.2 K in the Second Assessment Report (IPCC 1996). One essential factor in the uncertainty in climate sensitivity is the differences in the simulated cloud feedback in the models (e.g., Cess et al. 1990). Change in cloud radiative-forcing (CRF) is a crucial factor of cloud feedback and significantly affects a climate system. CRF changes in response to external forcing are very complicated, and the mechanism is not yet well understood. Many studies have attempted to attribute CRF changes to various aspects of cloud changes; for example, areas of cloud cover, and convective activity associated with changes in atmospheric circulation, and the thermodynamic structure or the sea-surface temperature (SST) (Pierrehumbert 1995; Ram-

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anathan and Collins 1991), or changes in the cloud optical properties and hydrological cycle (Stephens 2005 and references therein), without providing a comprehensive explanation. One certain way to solve the problem is to improve the climate models so that they can realistically simulate present-day climate and variation for the CRF.

MRI-CGCM2.0 has been used in many climate studies (e.g., Yukimoto and Kitamura 2003) since its release, in which several issues concerning the model's performance have been identified. One factor is that the model's climate sensitivity is at a lower boundary among the models used for IPCC TAR assessments, though it is not necessarily incorrect, at least in the present consensus. The simulated global radiation budgets at the top of the atmosphere (TOA) for solar and longwave radiations differed considerably from the satellite observation (e.g., Earth Radiation Budget Experiment; ERBE) (Barkstrom et al. 1989), although the bias in the net component was smaller, and the simulated climatological CRF did not agree with the estimation from the ERBE (Harrison et al. 1990).

Several modifications have been made in the new version of the model, particularly in the cloud scheme, for the purpose of improving the general performance of the model in reproducing the present-day climate. We evaluate here how the model achieves that objective by comparing the simulated present-day climate with that by the previous version and observations for various statistics, such as the global energy budget, meridional distribution of the radiation balance, and the geographical and seasonal variation of the fields of the atmosphere and the oceans (sea ice). We also evaluate the climate sensitivity of the model and determine the most essential factor in the climate-sensitivity difference between the two versions, as well as how the change in the cloud response contributes to the difference.

Many simulations produced by the new version, such as paleoclimate simulations, historical climate simulations and scenario experiments for projection of future climate changes, were performed for our contributions to the IPCC Fourth Assessment Report (AR4). Yukimoto et al. (2005) imposed historical forcing from natural and anthropogenic sources to simulate the climate change in the twentieth century. They provided simulated climatic trends in the late twentieth century for various fields, including the global surface-air temperature (SAT) and atmospheric circulations, with good agreement with those in observed climate changes. They also performed experiments to project future climate changes based on the IPCC scenario. A portion of the simulation results were provided for lower-boundary conditions (the SST and sea-ice distribution), in the time-slice climate change experiments (Oouchi et al. 2006) with a very high-resolution (20-km mesh) atmospheric GCM (Mizuta et al. 2006). This paper offers important information about the basic climate state and the climate sensitivity of the model that is necessary to interpret and apply the results of those studies.

A description of the model that focuses on modifications relative to the previous version (MRI-CGCM2.0) is provided in the following section. Descriptions of the experiments for evaluating the present-day climate and climate sensitivity are then presented in Section 3. The present-day climate, simulated by the new version is described in Section 4, with comparisons with results from the previous version. Changes in climate sensitivity and their association with cloud-forcing changes are discussed in Section 5. Conclusions are given in Section 6.

2. Description of the model

The new version of the model is referred to as MRI-CGCM2.3 (hereafter MRI2.3). The fundamental framework of this version is the same as the previous version MRI-CGCM2.0 (Yukimoto et al. 2001; hereafter MRI2.0). The following model description is applicable to both versions, except for the modifications summarized in Table 1. The model consists of an atmospheric general circulation model (AGCM), and a global ocean general-circulation model (OGCM), which include respective sub-models of land processes and sea ice. We provide a detailed description of the cloud scheme, because it plays an important role in determining the climate sensitivity of the model.

2.1 Atmospheric component

The atmospheric component is fundamentally identical to the AGCM MRI/JMA98 (Shibata et al. 1999) developed at MRI, and the

Table 1. List of modifications from MRI-CGCM2.0 to MRI-CGCM2.3

Model Version	MRI-CGCM2.0	MRI-CGCM2.3
Critical relative humidity R_c in cloud scheme	Function of pressure level	Function of pressure level separately determined for convective or layer cloud, and over ocean or land
Diagnosis of tropopause height for cloud scheme	Adiabatic lapse rate	Adiabatic lapse rate + temperature
Effective radius of cloud droplet	10.0 μm	8.4 μm
Ocean-surface albedo α_{ocn} (function of solar zenith angle μ)	$\alpha_{ocn} = 0.12347 + 0.34667\mu - 1.7485\mu^2 + 2.04630\mu^3 - 0.74839\mu^4$	$\alpha_{ocn} = 0.026/(\mu^{1.7} + 0.065) + 0.15 \cdot (\mu - 0.1)(\mu - 0.5)(\mu - 1.0)$: for beam part $\alpha_{ocn} = 0.06$: for diffused part (Briegleb et al. 1986)
Land-surface albedo	Default values	Default values + 0.04
Ocean diapycnal mixing coefficient	1×10^{-5} m^2 s^{-1} (constant)	Varying with depth (Tsujino et al. 2000)
Spin up for the control simulation	Asynchronous coupled 209 years + synchronous coupled 111 years	Synchronous coupled 429 years

same modifications were made in its updated versions. The dynamic framework is a spectral transform method, with a horizontal resolution of T42 in wave truncation, and 64×128 (about 2.8×2.8 degrees grid spacing in latitude and longitude) in a transformed Gaussian grid. The vertical configuration consists of a 30-layer sigma-pressure hybrid coordinate, with the top at 0.4 hPa. The velocity potential and streamfunction (equivalent to zonal and meridional velocities), temperature and specific humidity are prognostic variables, and cloud water and cloud amount are treated as diagnostic variables.

The model uses a multi-parameter random model based on Shibata and Aoki (l989) for terrestrial radiation calculation. The model treats four spectral bands (wave-number intervals are 20 to 550, 550 to 800, 800 to 1200, and 1200 to 2200 cm^{-1}). It explicitly addresses absorption due to methane (CH_4) and nitrous oxide (N_2O), in addition to water vapor (H_2O), carbon dioxide (CO_2), and ozone (O_3). The delta-two-stream approximation is used for solar radiation calculation. The spectral intervals, the data for k-distribution, absorption cross sections, and optical constants of clouds were taken from Briegleb (1992); these take into account O_3 in the ultraviolet and visible regions (eight intervals between 0.2 and 0.7 μm), H_2O in the near-infrared region (seven intervals between 0.5 and 5.0 μm), CO_2 (2.7 and 4.3 μm), and O_2 (A and B bands). The delta-two-stream calculation for transmission and reflection is performed using the discrete ordinate method (Shibata and Uchiyama 1992). Scattering of solar radiation by atmospheric aerosols (direct effect) is explicitly treated. The optical properties of sulfate aerosol are substituted by those for a rural aerosol model (Shettle and Fenn 1979) in which the aerosols are assumed to be composed of a mixture of 70% water-soluble substances (ammonium sulfate, calcium sulfate, and organic compounds), and 30% dust-like aerosols. These parameters are applicable for sulfate aerosols, since the complex refractive indices for water-soluble and dust-like aerosols are very similar, particularly in the solar wavelength region.

We used the Arakawa-Schubert scheme (Arakawa and Schubert 1974) with prognostic closure similar to that of Randall and Pan (1993) for deep, moist convection. A level-2 turbulence closure scheme, based on Mellor and Yamada (1974), was used for vertical diffusion. Oro-

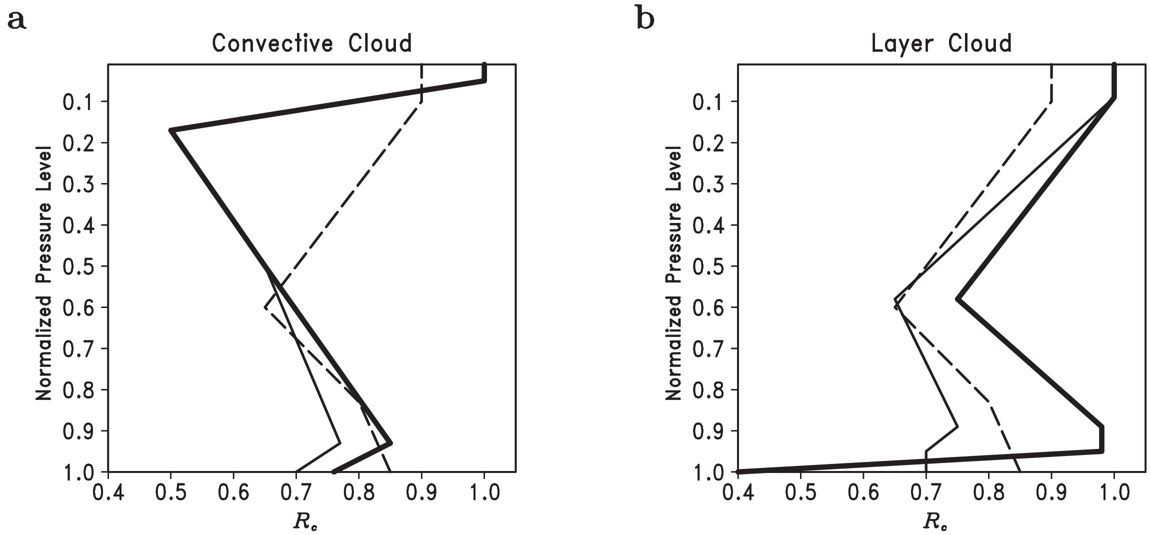


Fig. 1. Vertical profiles of critical relative humidity R_c with respect to σ to diagnose cloud fractions for (a) convective clouds and (b) layer clouds. The solid lines denote the profiles used in MRI2.3 for clouds over ocean (bold) and over land (thin). The dashed lines denote the profile used in MRI2.0 for all clouds (the same profile is used for both convective and layer clouds in MRI2.0).

graphic gravity-wave drag was parameterized by a scheme developed by Iwasaki et al. (1989), and we used Reyleigh friction for non-orographic gravity-wave drag.

The parameter for ocean-surface albedo was modified in MRI2.3 from that in MRI2.0 (Table 1). It is determined by a function of the solar zenith angle based on Briegleb et al. (1986), which indicates a significantly higher value for large zenith angles when compared with the function used in MRI2.0. In addition, the surface albedo over land is uniformly increased by 0.04. These modifications make the clear-sky reflected shortwave radiation greater in MRI2.3 than in MRI2.0.

a. Cloud scheme

The model treats clouds with a diagnostic cloud scheme. The cloud fraction, f_c, is calculated from relative humidity R as:

$$f_c(\sigma) = \begin{cases} \left[\dfrac{R - R_c(\sigma)}{1 - R_c(\sigma)}\right]^2 & \text{for } R \geq R_c(\sigma) \\ 0 & \text{for } R < R_c(\sigma) \end{cases},$$

where R_c is the critical relative humidity required to form a cloud, and σ is a pressure level normalized by the pressure difference between the tropopause and the surface. Critical relative humidity R_c was a function of the pressure level alone in MRI2.0. However, the relationship between cloud amount and relative humidity reflects not only the cloud-microphysics processes, but also reflects the sub-grid scale inhomogeneity of water vapor in a grid cell. It is difficult to determine with a single function of relative humidity and height. We modified this to use different profiles of R_c separately for convective clouds and for layer clouds. Furthermore, different profiles are applied for clouds over ocean and clouds over land, which can roughly express the effect from a difference in condensation-nuclei density between ocean and land. We optimized the profiles (as illustrated in Fig. 1), so that the simulated distribution of the radiation at TOA, resembles the ERBE observation. The tropopause height was overestimated in the winter high latitudes by MRI2.0, with its procedure based on the adiabatic lapse rate alone. The tropopause height is limited in MRI2.3 to below a level, with the temperature at 226 K. The cloud fraction, based on the above relative-humidity function, is reduced over a shallow convection area by a factor depending on the static stability, as in MRI2.0.

The cloud water content, w, is estimated from temperature T using an empirical relationship (Stephens et al. 1990):

$$w = 0.3 \times 10^{0.44T}$$

where w is in g m^{-3}, and T is in °C. A minimum temperature of about −43°C was introduced to retain cloud water content that exceeds a certain value.

Cloud emissivity in terrestrial radiation is approximated to be independent of wavelength and direction as:

$$\varepsilon = 1 - exp(-1.66\alpha W_c),$$

where W_c is the cloud water path in g m^{-2}, and the absorption coefficient α is 0.08, which is nearly an arithmetic average of those for upward and downward directions (Stephens 1978). Cloud overlap is calculated using a correlation coefficient, so that it resembles the maximum overlap between near level clouds, but becomes a random overlap between distant level clouds.

A partial cloud is extended to form an overcast cloud in the solar radiation calculation, following Briegleb (1992). Its optical parameters are defined similarly with the formulation by Slingo (1989). The effective radius of cloud droplets is assumed to be a constant of 8.4 μm, which is smaller than the 10.0 μm in MRI2.0. Cloud droplets have considerable diversity in their properties (including their size distribution); each contains great uncertainty, and geographical and temporal variations. These cloud properties are closely related with the aerosol indirect effect (first kind), which is an important factor in climate change, as well as the aerosol direct effect. However, our model does not incorporate the indirect effect. We adopted a strategy in which we place emphasis on the overall balance in the model's climate, rather than sophisticating the cloud physics. Therefore, the effective radius is expressed by one parameter, and it is adjusted so that the shortwave cloud forcing comes close to the observation in the present-day simulation, as described later.

2.2 Land component

The land component is based on the simple biosphere (SiB) model (Sellers et al. 1986; Sato et al. 1989), which includes the effects of vegetation. The land model has three soil layers with different field capacities depending on the vegetation type, in which the temperature, liquid water, and frozen water in each soil layer are predicted. Canopy and grass are treated for each of the 13 vegetation types, for which the parameters are dependent on the vegetation type and month of the year. The model strictly conserves energy and water mass in the phase transitions (melting and freezing) of soil water. This is important for evaluating changes in snow cover and permafrost in climate changes. Snow cover is also predicted as one bulk layer, with a surface skin layer, by accounting for thermodynamical processes. The surface albedo over the land depends on the vegetation type for soil skin, and also depends on the snow-skin temperature, T_{snow}, if it is covered by snow. The snow-skin albedo ranges from 0.8 ($T_{snow} < -4$°C) to 0.64 ($T_{snow} = 0$°C, melting snow) for visible bands, and from 0.4 ($T_{snow} < -4$°C) to 0.32 ($T_{snow} = 0$°C, melting snow) for near infrared bands.

Runoff from the soil layers (surface runoff plus underground runoff) is transferred to either a river-mouth grid or an inland-water grid, followed by river routing. When accumulated snow over the ice sheets exceeds a threshold water-equivalent depth (10 m), the excess water is treated the same as runoff, and discharged immediately (through the glacier routing determined based on surface altitudes) to the ocean as icebergs. Five lakes, i.e., the Caspian Sea, the Aral Sea, Lake Balkhash (central Asia), Lake Chad (west central Africa), and Lake Eyre (Australia), were modeled as inland waters. We considered fractional coverage of inland water in a grid box. Storage of inland water changes through a balance between river discharge, precipitation, and evaporation over the water area. The water temperature is predicted by the heat budget at the water surface, assuming a slab with a thickness of 50 m (independent of water storage).

2.3 Ocean component

The oceanic component of the model is a Bryan-Cox-type ocean general-circulation model (OGCM) with a global domain. The horizontal grid spacing is 2.5 degrees in longitude and 2 degrees in latitude poleward of 12 degrees in both hemispheres. The meridional grid spacing is set at 0.5 degrees near the equator between latitudes 4°S and 4°N to obtain good resolution for the equatorial oceanic waves. The grid spacing gradually increases from 0.5 de-

Table 2. Vertical configuration of the ocean component in the MRI-CGCM2.3

Level	Depth (m)	Thickness (m)	Diapycnal mixing coefficient (10^{-5} m^2s^{-1})
1	5.2	5.2	1.328
2	11.4	6.2	1.334
3	19.3	7.9	1.341
4	30.0	10.7	1.350
5	55.0	25.0	1.374
6	80.0	25.0	1.399
7	105.0	25.0	1.426
8	130.0	25.0	1.455
9	155.0	25.0	1.485
10	180.0	25.0	1.517
11	210.0	30.0	1.559
12	250.0	40.0	1.620
13	300.0	50.0	1.705
14	400.0	100.0	1.910
15	600.0	200.0	2.497
16	800.0	200.0	3.411
17	1100.0	300.0	5.609
18	1500.0	400.0	10.664
19	2200.0	700.0	15.137
20	2900.0	700.0	20.274
21	3600.0	700.0	23.291
22	4300.0	700.0	25.051
23	5000.0	700.0	26.003

grees to 2 degrees for latitudes between 4 and 12 degrees. The model has 23 levels unevenly spaced in the vertical direction, as indicated in Table 2. The uppermost layer has a thickness of 5.2 m, and the deepest bottom is set to 5000 m. The Denmark Strait is made slightly deeper and broader than the real topography, to represent the subgrid-scale overflow of waters that form in the Nordic Seas. This modification contributes to the realistic thermohaline circulation in the North Atlantic.

Parameterized subgrid-mixing processes using viscosities and diffusivities are specified as follows. The horizontal-viscosity coefficient is 1.6×10^5 m^2s^{-1}, and the vertical-viscosity coefficient is 1×10^{-4} m^2s^{-1}. An eddy-mixing parameterization based on Gent and McWilliams (1990) is introduced, wherein the isopycnal-mixing coefficient is set uniformly to 2×10^3 m^2s^{-1}. The diapycnal-mixing coefficient has been modified in MRI2.3 to vary with depth, based on Tsujino et al. (2000) as provided in Table 2; it was a constant 1×10^{-5} m^2s^{-1} in MRI2.0. The vertical turbulence viscosity and diffusivity, were modeled following Mellor and Yamada (1974, 1982), and Mellor and Durbin (1975) for the surface mixed layer, in addition to the isopycnal mixing. A convective adjustment that mixes the whole vertical column is applied when vertical stratification becomes unstable. Solar radiation that penetrates the seawater is absorbed with an *e*-folding depth of 10 m, which heats several dozen meters of surface seawater.

2.4 *Sea-ice component*

The sea-ice component is similar to the model developed by Mellor and Kantha (1989). Compactness and thickness are predicted based on the thermodynamics and the horizontal advection and diffusion. The freezing and melting rates of sea ice are calculated with balances of heat and fresh water at the sea-ice bottom, at the open sea surface (freezing only), and within seawater (creation of frazil ice).

The compactness calculation contains empirical constants as the tuning parameters. The sea-ice volume is advected by the surface ocean current multiplied by an empirical constant (set to 1/3). The compactness is also advected by the surface ocean current, with maximum limits of 0.997 in the northern hemisphere, and 0.98 in the southern hemisphere. Horizontal diffusions are applied for both thickness and compactness, with a constant diffusivity of 5×10^3 m^2s^{-1} in addition to the advection.

The surface albedo on the sea ice is calculated dependent on whether it is snow-covered, and on its skin temperature, T_{skin}. The surface albedo ranges from 0.85 ($T_{skin} < -3°C$) to 0.7 ($T_{skin} = 0°C$, melting snow) for snow-covered sea ice. The surface albedo ranges from 0.75 ($T_{skin} < -6°C$) to 0.55 ($T_{skin} > -2°C$) for bare sea ice.

2.5 *Coupling scheme*

The atmosphere and the ocean interact by exchanging fluxes of heat, fresh water, and momentum at the sea surface, and through the SST and sea-ice distribution. The contents of the fluxes are sensible- and latent-heat fluxes, net shortwave and longwave radiation, precipitation and evaporation, river discharge and melting water from snow and ice over the sea ice, and zonal and meridional components of the surface wind stress.

The atmospheric component calculates the surface fluxes separately for underlying land, lake, ocean, and sea ice that may coexist in an atmospheric grid cell, which (for ocean and sea ice) is averaged for the 24-hour coupling interval. The fluxes are then transferred to the ocean and sea-ice components, which rigorously conserves the energy and water budgets.

2.6 *Spin-up and flux adjustment*

The model climatology of the present-day climate simulation is kept close to the observed conditions by using the flux adjustments for heat and fresh water obtained from the last part of the 334-year calibration run, in which the surface temperature and salinity have been restored to the observed climatology (Levitus et al. 1994; Levitus and Boyer 1994). We also adjusted for wind stress only in the equatorial region to reproduce a realistic climatological thermocline along the equator, the structure of which plays an important role in the model performance in simulating tropical variabilities, such as El Niño.

The overall patterns of the flux adjustments are similar to those for MRI2.0 (see Fig. 1 of Yukimoto et al. 2001); however, the globally averaged heat-flux adjustment results in a much smaller value than that for MRI2.0. This is because the radiation budget is significantly improved, as described later.

A 95-year spin-up run, starting from the final state of the calibration run with applying the flux adjustments, is performed to make the initial condition suitable for the control simulation with a small drift. We used an asynchronous coupled integration for the spin-up for the MRI2.0 simulations to save computational time. However, we used synchronous (normal) coupling for the MRI2.3 spin-up, since switching from asynchronous to normal coupling requires additional long integration to stabilize the model.

2.7 *Slab ocean model*

The model can be used as a slab ocean-coupled GCM (SGCM) by replacing the ocean component with a 50-m-thick thermodynamic mixed-layer ocean. The sea-ice component of the SGCM is identical to that in the full CGCM, except that no advection is applied. The temperature of the slab is maintained close to climatological SST by the use of a monthly varying *q*-flux, which is obtained from a calibration run similar to that for a heat-flux adjustment in the full CGCM.

3. Experiments

In this section, we describe the experiment design used exclusively to evaluate the control climate, and climate sensitivity of the model; other experiments (such as historical and scenario experiments) for the IPCC AR4, are described in Yukimoto et al. (2005).

We performed a present-day control (referred to as PDcntrl) simulation with a 500-year duration, with external forcings fixed at the present-day levels, to evaluate the control climate of MRI2.3. We set the concentration of atmospheric CO_2 at 348 ppmv; the same setting was imposed for all the other forcings during all the spin-up runs described above. The initial state of the PDcntrl simulation was taken from the final state of the spin-up run. A similar present-day control simulation was also prepared using MRI2.0 for comparison, with the same settings except for the sulfate aerosol (referenced later), and with a 150-year duration.

We generated two 500-year-long CGCM simulations to investigate the transient climate response (TCR) of MRI2.3. The CO_2 concentration in these simulations was increased at a rate of 1%/year compound until it was doubled (at year 70) for one simulation (referred to as 1%to2x), and quadrupled (at year 140) for the other simulation (referred to as 1%to4x), and was stabilized thereafter. Other forcing agents were set the same as in the PDcntrl. The initial states for the 1%to4x, and 1%to2x simulations, were taken from the first and 51st years of the PDcntrl simulation. A similar TCR experiment was performed with MRI2.0, but only for 1%to4x and with a 150-year duration.

We also prepared a set of SGCM simulations (150 years) for a control, and doubled the CO_2 (696 ppmv) to examine the equilibrium climate sensitivity. These simulations are referred to as Slabcntl and 2xCO2. The forcing agents, except for the CO_2 concentration, were the same as in the PDcntrl simulation with the CGCM. These SGCM simulations are also part of the experiments for the IPCC AR4 contribution.

The external forcing agents that can be imposed on the MRI-CGCM2 include greenhouse gases, sulfate aerosol, solar activity, and volcanic activity. Concentrations of the greenhouse gases, except for CO_2, were set constant at 1.650 ppmv for CH_4, and 0.306 ppmv for N_2O as the present-day levels for all the simulations.

The sulfate aerosol was specified by the monthly varying horizontal distribution of the column total mass. Sulfate aerosol from natural sources was included in the MRI2.3 simulations, in addition to anthropogenic sulfate, with the mass loading data for both origins based on estimation by a chemical transport model (Tegen et al. 1997), available at http://gacp.giss.nasa.gov/transport/. Only the present-day anthropogenic aerosol distribution (Langner and Rodhe 1991) was imposed in the MRI2.0 simulations. The mixing ratio of the sulfate aerosol was uniformly vertically distributed within the lower layers below 740 m in the MRI2.3 simulations, and below 1600 m in the MRI2.0 simulations. This modification was introduced to reduce the discontinuity of the forcing between the present-day and scenario experiments (Yukimoto et al. 2005), for which the three-dimensional sulfate aerosol distributions were obtained from simulations with another chemical transport model.

Forcing by solar activity can be simply imposed by changing the solar constant, but it was fixed at 1367 Wm^{-2} for the PDcntrl and TCR simulations. Forcing by volcanic activity is due to radiative forcing of aerosols injected into the stratosphere by intense volcanic eruptions. Volcanic forcing was not included in the simulations in the present study, though it could be included (for such purposes as historical simulations) by changing the solar constant with equivalent radiative forcing.

4. Control climate

4.1 Climatic stability of the control simulation

It is important that the atmosphere and oceans be free from any climatic drifts in a climate model used for climate-change simulations. The globally averaged annual mean SAT for the PDcntrl simulation (Fig. 2), demonstrated very stable behavior over the 500 years. The linear trend of the time-series over the 500 years was +0.0027 K/century, effectively indicating no temperature trend in the control simulation compared to the magnitude of the trend in the TCR experiments (presented later).

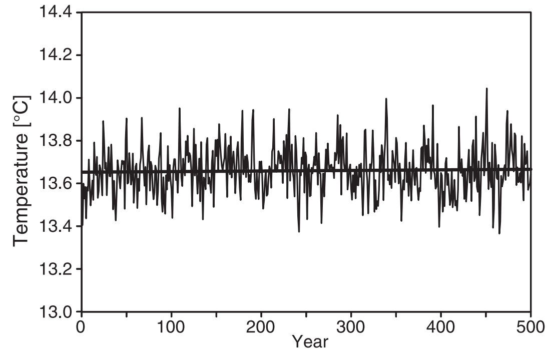


Fig. 2. Temporal variation of the globally averaged annual mean SAT (unit is °C) in the PDcntrl simulation by MRI2.3 and its linear trend (thick line).

Climatic stability of the sea ice in a control simulation is very important, since sea ice interacts strongly with both the atmosphere and oceans, despite its relatively small coverage area on a global scale. The annual mean sea-ice areas (Fig. 3), displayed essentially no climatic drift for the northern hemisphere, while there was a moderate decreasing trend in the first one and a half centuries, followed by more stable behavior with periodic centennial-scale variation in the southern hemisphere (compared to the northern hemisphere). The simu-

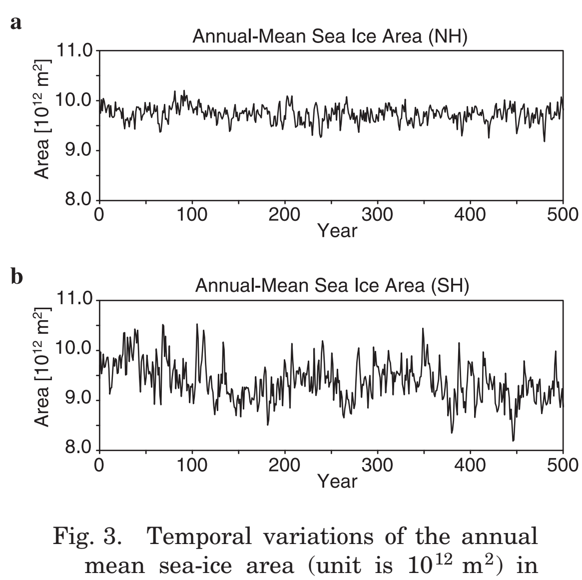


Fig. 3. Temporal variations of the annual mean sea-ice area (unit is 10^{12} m^2) in (a) the northern hemisphere and (b) the southern hemisphere in the PDcntrl simulation by MRI2.3.

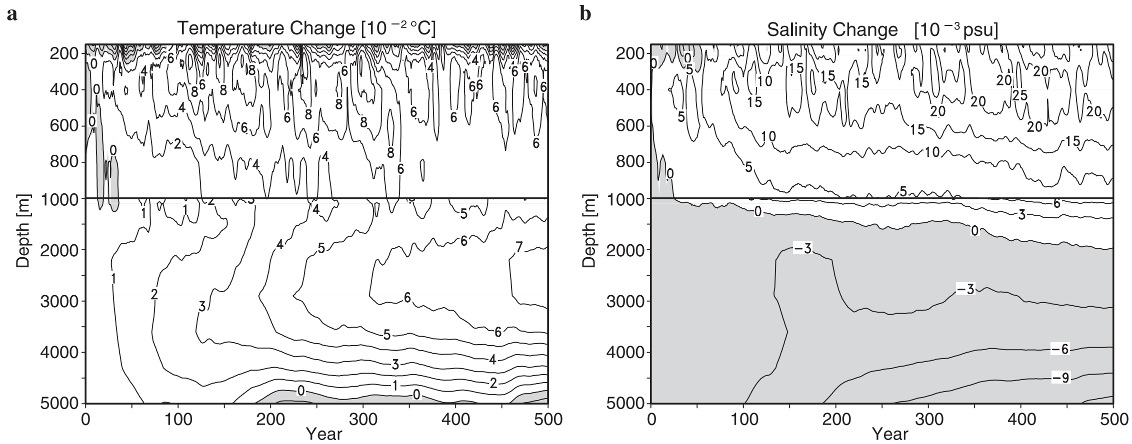


Fig. 4. Temporal variation of the oceanic drifts (deviations from the initial condition) in the PDcntrl simulation by MRI2.3 for (a) potential temperature (unit is 10^{-2}°C) and (b) salinity (unit is 10^{-3} psu), averaged horizontally over the global ocean domain.

lated annual mean areas were comparable with the observed values (HadISST-1.1, Rayner et al. 2003) of 10.7×10^{12} m^2 for the northern hemisphere, and 9.8×10^{12} m^2 for the southern hemisphere, though they were slightly underestimated in both hemispheres. In addition to the annual mean areas, the seasonal variation and geographical distribution (not shown) were also stably simulated with sufficient agreement with the observations.

Figure 4 illustrates the temporal changes of the horizontally averaged potential temperature and salinity for the global ocean. Both the temperature and salinity exhibited sufficiently small drifts at all depths. There were essentially no drifts in either temperature or salinity in the surface layer, except for the first several decades. In contrast, there was a relatively greater temperature trend in the deep layers at around 2500 m, though the magnitude of 0.03°C/century was sufficiently small. The model revealed salinity drifts, with an increasing trend in the upper layers with a maximum of 0.015 psu/century around a depth of 400 m, and a decreasing trend in the deepest layers by -0.002 psu/century. These drifts were much smaller than those in MRI2.0; their size limited us to evaluating transient climate responses within a time scale of, at most, a few centuries. The meridional overturning of the oceans (not shown) also revealed the ample stability in the control simulation by MRI2.3. For instance, the temporal variation of the maximum overturning in the North Atlantic indicated a very small linear trend (0.12 Sv/century) at around 20 Sv, relative to the standard deviation of 0.8 Sv for the interannual variation.

The PDcntrl simulation by MRI2.3, exhibited excellent stability and essentially no climatic drift, at least in the atmosphere and the upper oceans. We describe the fundamental performance of the model in the remainder of this section, by examining the simulated control climate in the PDcntrl. We focus in particular on the aspects associated with climate sensitivity and the modifications of MRI2.0, such as global energy balance, cloud distribution, and CRF. Simulated data are provided for the averages over 30 years, from 61 to 90 years of PDcntrl simulations by MRI2.3 and MRI2.0. Both versions exhibited minimal climatic drift in the surface temperature during that period.

4.2 *Energy balance*

The global averages of the simulated radiation budget at TOA are indicated in Table 3, together with the ERBE observations. The ERBE observations are averages from February 1985 to January 1988. The outgoing longwave radiation (OLR) (236 Wm^{-2}) was close to the observation, while it was underestimated (214 Wm^{-2}) in MRI2.0. Although the reflected shortwave (SW) radiation decreased and became smaller than the observation, down welling shortwave at the surface became greater than that in

Table 3. Globally averaged annual-mean fluxes at the TOA and surface and cloud forcings at the TOA for the models' control climate and the observations

		MRI2.0	MRI2.3	Observations
Reflected SW at TOA	[Wm^{-2}]	113	102	107*1
Outgoing LW at TOA	[Wm^{-2}]	214	236	235*1
Net radiation budget at TOA	[Wm^{-2}]	15	4	0
Clear-sky reflected SW at TOA	[Wm^{-2}]	41	53	56*1
Clear-sky outgoing LW at TOA	[Wm^{-2}]	261	258	264*1
Downwelling SW at surface	[Wm^{-2}]	174	197	198*2
Downwelling LW at surface	[Wm^{-2}]	351	337	324*2
Net down. SW at surface	[Wm^{-2}]	154	165	168*2
Net upward LW at surface	[Wm^{-2}]	55	59	66*2
Sensible heat flux at surface	[Wm^{-2}]	18	31	24*2
Latent heat flux at surface	[Wm^{-2}]	68	75	78*2
SW cloud forcing at TOA	[Wm^{-2}]	−72	−49	−49*1
LW cloud forcing at TOA	[Wm^{-2}]	47	22	30*1
Net cloud forcing at TOA	[Wm^{-2}]	−25	−27	−19*1
Globally averaged precipitation	[mm day^{-1}]	2.42	2.60	2.62*3

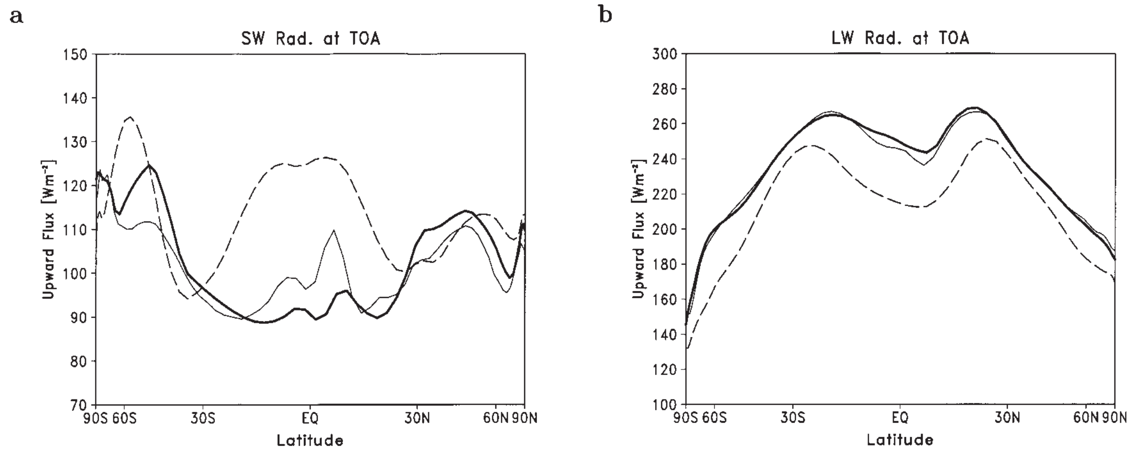
*1: ERBE, *2: Kiehl and Trenberth (1997), *3: GPCP-V2 (average over 70 S–85 N)

MRI2.0 and was closer to the observation. The net (LW + SW) radiation budget of 4 Wm^{-2} in MRI2.3, exhibited much better balance compared to the overestimated heating of 15 Wm^{-2} in MRI2.0. The clear-sky reflected (upward) shortwave at TOA significantly increased by about 12 Wm^{-2}, which can be primarily attributed to the modified ocean-surface albedo (about 8 Wm^{-2} estimated from a one-year impact test). The increased land-surface albedo also contributed about 2 Wm^{-2}. Scattering by the inclusion of natural source aerosols, modification of the vertical distribution of the aerosols, and the use of different data for the anthropogenic sulfate aerosol contributed a smaller part.

Overall, the energy budgets at the surface were also satisfactorily simulated by MRI2.3. The shortwave- and longwave-radiation budgets at the surface approached the observational estimation by Kiehl and Trenberth (1997). Both sensible- and latent-heat fluxes at the surface were increased in MRI2.3. As a result, the globally averaged precipitation approached the observation, which implies a reasonable hydrological cycle, as well as energy transport.

The radiation budgets at TOA are examined in the zonal mean meridional distribution (Fig. 5). MRI2.0 significantly overestimated the upward shortwave radiation in the tropics. It was remarkably reduced in MRI2.3 and was closer to the observation, though there was an underestimation in the Inter-Tropical Convergence Zone (ITCZ). The longwave radiation was underestimated by MRI2.0 at all latitudes. It was generally increased in MRI2.3, and became satisfactorily close to the observation at all latitudes except near the ITCZ, where it was slightly overestimated. The biases near the ITCZ in both shortwave and longwave appeared to be associated with a lack of anvil clouds of cumulus convection. The meridional distribution of the net radiation budget for MRI2.3 exhibited generally better representation than for MRI2.0, except for a slight overestimation of the radiation input at the tropics. The meridional distribution for each month (not shown) was also improved, as well as for the annual mean.

The accurate radiation budget in the meridional distribution is expected to lead to more realistic meridional heat-transport in both the atmosphere and the ocean. The implied heat transport by the global ocean was calculated by integrating the zonal-integrated surface-heat flux (including flux adjustment) southward from the North Pole. The heat transports in the PDcntrl simulations are illustrated in Fig. 6. By definition, there must be a tempera-



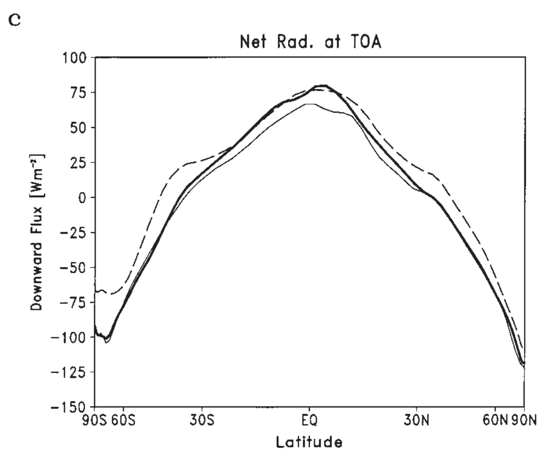


Fig. 5. Meridional distributions of the zonally averaged annual mean radiative fluxes at TOA for the (a) upward shortwave, (b) upward longwave, and (c) downward net radiation in the PDcntrl simulations by MRI2.3 (thick solid) and MRI2.0 (dashed) and for the ERBE observation (thin solid).

ture drift in the ocean interior if the value at the South Pole is not equal to zero. The value at the South Pole was almost zero for MRI2.3, indicating a very small temperature drift in the ocean. The estimation excluding the flux adjustment is also depicted, and exhibits nearly zero (-0.1 PW) at the South Pole, which implies that the temperature drift would be sufficiently small as well, even without the flux adjustment. In contrast, there was a large discrepancy of -6.5 PW at the South Pole in the estimation by MRI2.0 without the flux adjustment (not shown). The implied heat transport by the ocean in MRI2.3 was 1.8 PW at 24°N, and -1.0 PW at 20°S. These values are consistent with the estimation from hydrographical observations (e.g., Ganachaud and Wunsch 2000).

4.3 Surface-air temperature

Figure 7 illustrates the geographical distributions of the SAT bias (deviation from the observation by Jones et al. 1999) for the northern winter (December to February; DJF), and northern summer (June to August; JJA). The MRI2.0 simulation displayed warm biases in the high latitudes over the Eurasian and North American continents in winter (Fig. 7a), which were alleviated in MRI2.3 (Fig. 7b). A warm bias in MRI2.0 over the winter Arctic Ocean, covered by sea ice, changed to a cold bias in MRI2.3. We consider these changes toward colder temperature in winter high latitudes to be possibly attributable to increased longwave cooling, related to cloud reduction. Some part of the change over the sea ice in winter is associated with the difference in sea-ice thickness between the two versions. MRI2.3 simulated thicker sea ice than MRI2.0 (3.75 m in MRI2.3 and 3.45 m in MRI2.0, for the average over 80°N to 90°N in DJF), which can lead to reduced heat conduction from the sea-ice bottom to the surface. However, it must be noted that the observed SAT also includes some uncer-

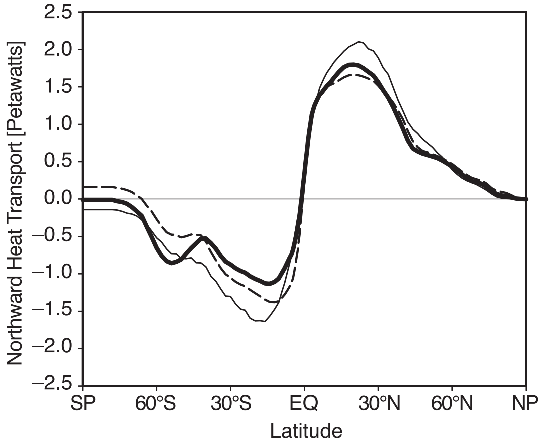


Fig. 6. Meridional ocean heat transport (unit is peta-watts, positive northward) in the PDcntrl simulations by MRI2.3 (thick solid) and MRI2.0 (dashed) implied by the surface heat flux and the tendency of ocean internal energy. The thin solid line denotes the implied ocean heat transport for MRI2.3 when the flux adjustment is excluded.

tainty in the Arctic and Antarctic regions, due to the very sparse observations.

MRI2.0 had an outstanding warm bias in summer, that prevails over most of the northern continents, with a peak value greater than 12°C (Fig. 7c). The warm bias in MRI2.3 was dramatically reduced (Fig. 7d). This reduction was associated with a reduction of solar radiation at the surface, due to an increase in the cloud amount over the land, which is primarily attributable to the modified cloud scheme, with different critical humidity between land and ocean. We infer that the reduction of solar radiation is balanced by the reduced sensible surface-heat flux, which leads to cooling of the air near the surface. The soil moisture increased with the decrease of surface temperature, leading to increases of evaporation and precipitation, which in turn resulted in positive feedback that increased the cloud amount.

The magnitudes of biases were generally reduced in MRI2.3. The overall reduction of the temperature bias reflects the regional and seasonal improvements, with individual reasons for each bias. The surface temperature over the oceans (except over sea ice) was close to the SST, and was also similar to the observation, since the simulated SST was maintained by successful flux adjustments.

4.4 Precipitation

The geographical distribution of precipitation simulated by MRI2.3 exhibited good agreement with the observed distributions (e.g., GPCP-V2, Adler et al. 2003) for DJF and JJA (Fig. 8) and other seasons (March to April, MAM; and September to November, SON; not shown) as well. The summer Asian monsoon and the Baiu in particular were simulated well, including their onset timing and migration (Rajendran et al. 2004), which are usually difficult to simulate correctly in coarse resolution models (e.g., Kang et al. 2002). However, some systematic drawbacks are commonly encountered in every season when compared with the observation (GPCP-V2). The simulated precipitation in the western Pacific through the eastern Indian Ocean was underestimated on the equator, and overestimated in the northern ITCZ, and around the South-Pacific Convergence Zone (SPCZ). The root-mean-square error (RMSE) of the biases for the region between 60°S and 80°N were 1.25 mm day^{-1} (DJF), 1.28 mm day^{-1} (MAM), 1.65 mm day^{-1} (JJA), and 1.38 mm day^{-1} (SON), generally reduced from those in MRI2.0 (1.35, 1.43, 1.57, and 1.42 mm day^{-1} for DJF, MAM, JJA, and SON), except for JJA.

The meridional distribution of the zonally averaged precipitation was also realistically simulated by the model (Fig. 9). The distributions in particular agree better with the observations in MRI2.3 than in MRI2.0 in both seasons (DJF and JJA). MRI2.0 tended to simulate double ITCZ, with a precipitation peak in the winter hemisphere, which was reduced in MRI2.3. However MRI2.3 simulated an overly strong northern ITCZ in JJA, that led to a slightly increased RMSE in JJA, as indicated above. MRI2.3 simulated more precipitation in the mid-latitudes in summer than MRI2.0, and its meridional distribution was closer to the observation.

Tropical precipitation distribution is primarily constrained by the balance between condensation heating and radiative cooling. Therefore, the realistic distributions of the radiative fluxes (which approached the observations, as indicated in Fig. 5) may possibly have lead to a

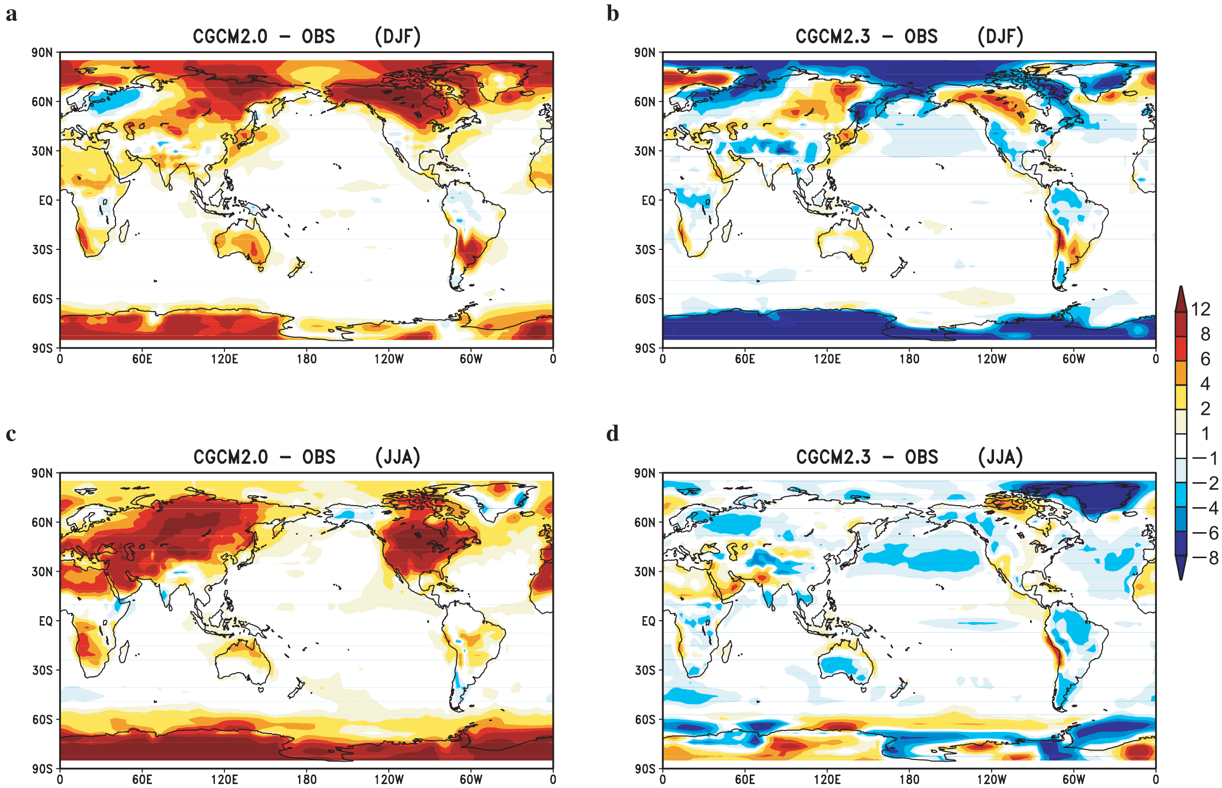


Fig. 7. Geographical distribution of the SAT deviation from the observation (Jones et al. 1999, average for 1961–1990) in the PDcntrl simulations by (a, c) MRI2.0 and (b, d) MRI2.3 for (a, b) DJF mean and (c, d) JJA mean.

more realistic precipitation distribution. The correct meridional distribution of precipitation also implies realistic meridional energy-transport in the atmosphere, both through sensible- and latent-heat transport, in conjunction with the correct meridional distribution of the radiation budget.

4.5 *Clouds*

We compared the meridional-vertical distributions of the simulated cloud-amount between MRI2.0 and MRI2.3 (Fig. 10). MRI2.0 simulated abundant low-level clouds in the tropical through subtropical latitudes, with a peak around the 900 hPa level. These low-level clouds were remarkably reduced in MRI2.3, which contributed predominantly to the reduced shortwave radiation bias in these latitudes (Fig. 5a), by reducing shortwave reflection. The tropical high clouds associated with cumulus convection were also reduced in MRI2.3, compared to MRI2.0, which led to the increase of OLR, that was underestimated in MRI2.0 and also partially contributed to reducing the overestimation of the shortwave reflection.

The cloud-top height in the extratropics was generally lowered in MRI2.3, particularly in the high latitudes in the southern hemisphere, where abundant clouds in the winter stratosphere were identified in MRI2.0. Mid-level clouds in the extratropics were reduced in MRI2.3, compared with MRI2.0, though the extratropical lowest-level clouds increased. These cloud-height changes contributed to diminishing the overall underestimation of the OLR in MRI2.0 (Fig. 5b).

These changes in cloud distributions are reflected in the distribution of CRF as viewed at TOA, which can be compared with the estimation from the observation (ERBE). The annual mean cloud-forcings for shortwave, longwave, and net radiations are illustrated in the meridional distributions of the zonal mean (Fig.

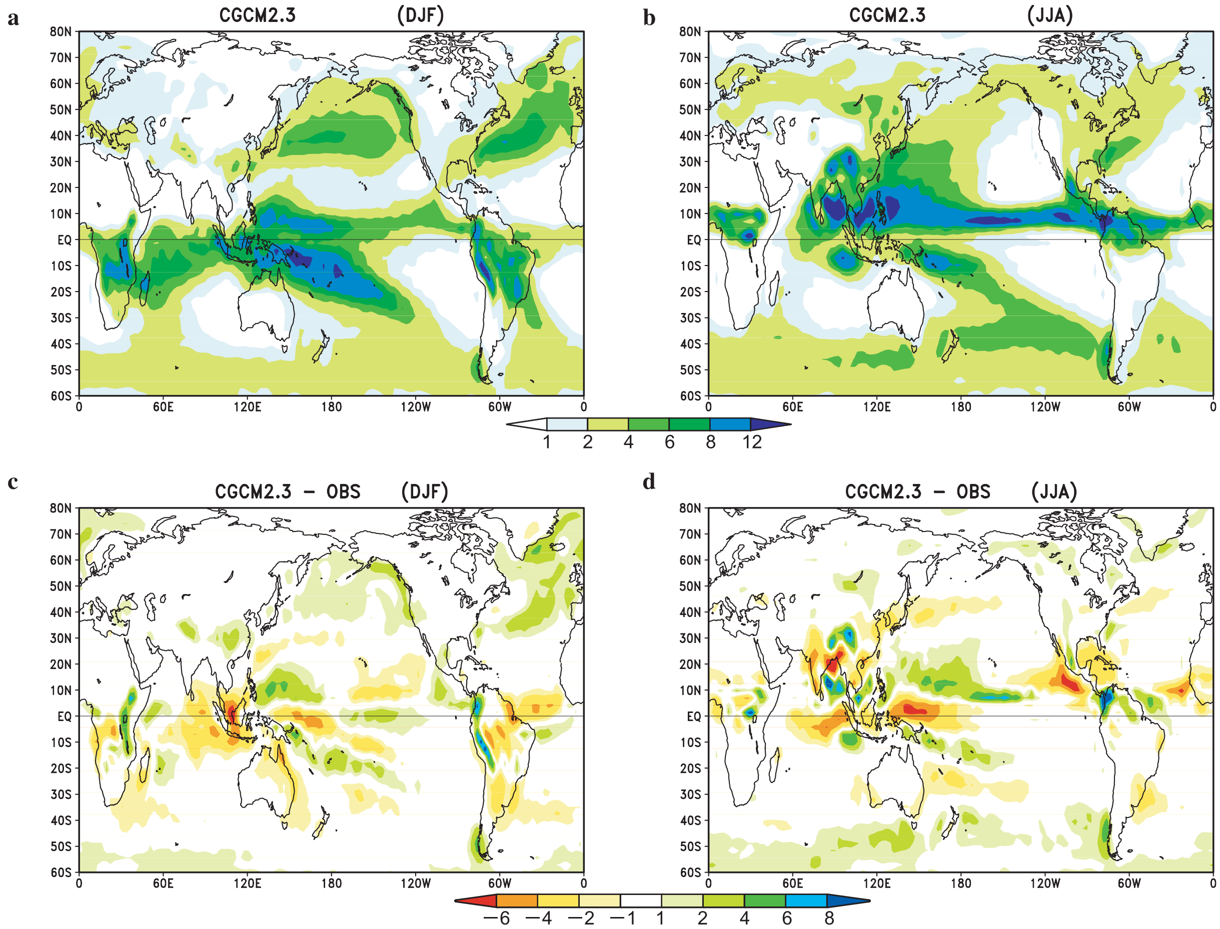


Fig. 8. Geographical distributions of the precipitation in the PDcntrl simulation by MRI2.3 for (a) DJF mean and (b) JJA mean, and its deviation from the observation (GPCP-V2, average for 1979–1999) for (c) DJF mean and (d) JJA mean.

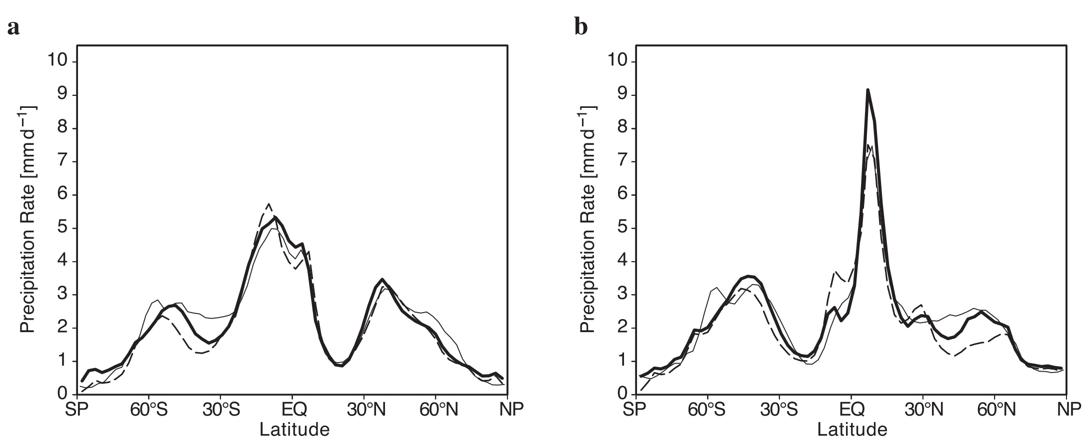


Fig. 9. Meridional distributions of the zonal mean precipitation for the (a) DJF mean and (b) JJA mean in the PDcntrl simulations by MRI2.3 (thick solid) and MRI2.0 (dashed) and that observed (thin solid, GPCP-V2, average for 1979–1999).

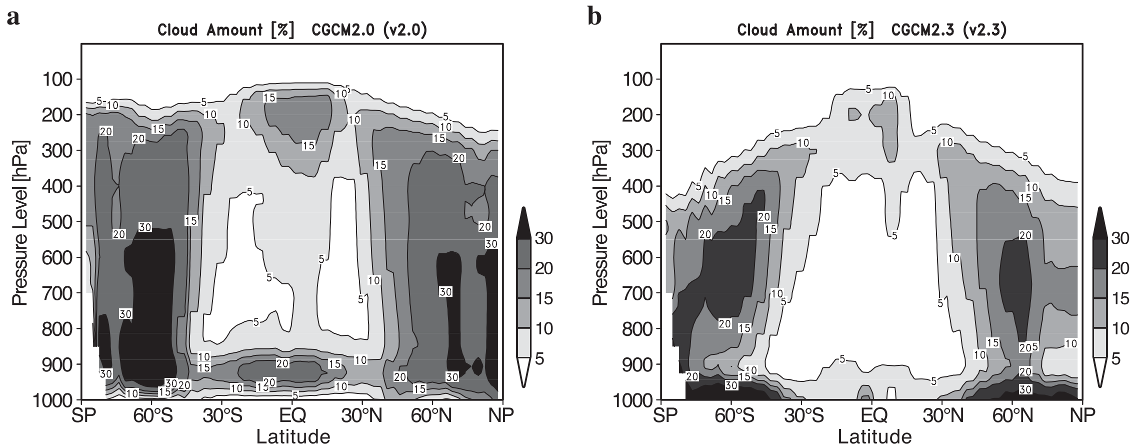


Fig. 10. Meridional-vertical distributions of the zonally averaged annual mean cloud-amount (%) in the PDcntrl simulations by (a) MRI2.0 and (b) MRI2.3.

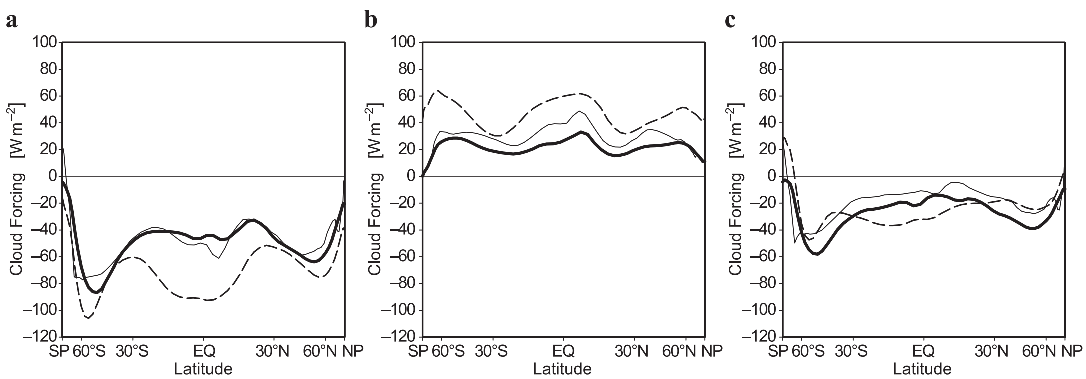


Fig. 11. Meridional distributions of the zonally averaged annual mean cloud forcing at TOA for (a) shortwave, (b) longwave, and (c) net radiations in the PDcntrl simulations by MRI2.0 (dashed) and MRI2.3 (thick solid) and the estimation from the ERBE observation (thin solid).

11), and in the geographical distributions (Fig. 12). Clouds generally create a negative forcing for shortwave radiation. MRI2.0 simulated much greater negative forcing (~ -90 Wm^{-2} in the zonal mean) in the tropics than in the observation, which was drastically reduced in MRI2.3 (~ -40 Wm^{-2} in the zonal mean), and approached the observation. The reduction was primarily associated with the reduced low-level clouds (Fig. 10) over the tropical oceans (Fig. 12). Negative biases in the extratropics in MRI2.0 were also generally reduced in MRI2.3, particularly over the continents in the northern high-latitudes (Fig. 12) associated with the decrease of cloud cover over land, due to the modified cloud scheme.

Clouds generally create a positive forcing for longwave radiation through their greenhouse effect. MRI2.0 exhibited an overall positive bias at all latitudes, with greater deviations from the observation in the high latitudes. In contrast, the longwave cloud forcing by MRI2.3 was decreased at all latitudes by about 20 to 40 Wm^{-2} in the zonal mean (Fig. 11b), and approached the observation in the high latitudes, though there was an underestimation (~ 10 Wm^{-2} in the zonal mean) in the low and middle latitudes, primarily for the high cloud regions (Fig. 12d). The reduction of longwave cloud forcing can be explained by the weakened greenhouse effect due to the lowered cloud-top height in MRI2.3. The net cloud forcing in

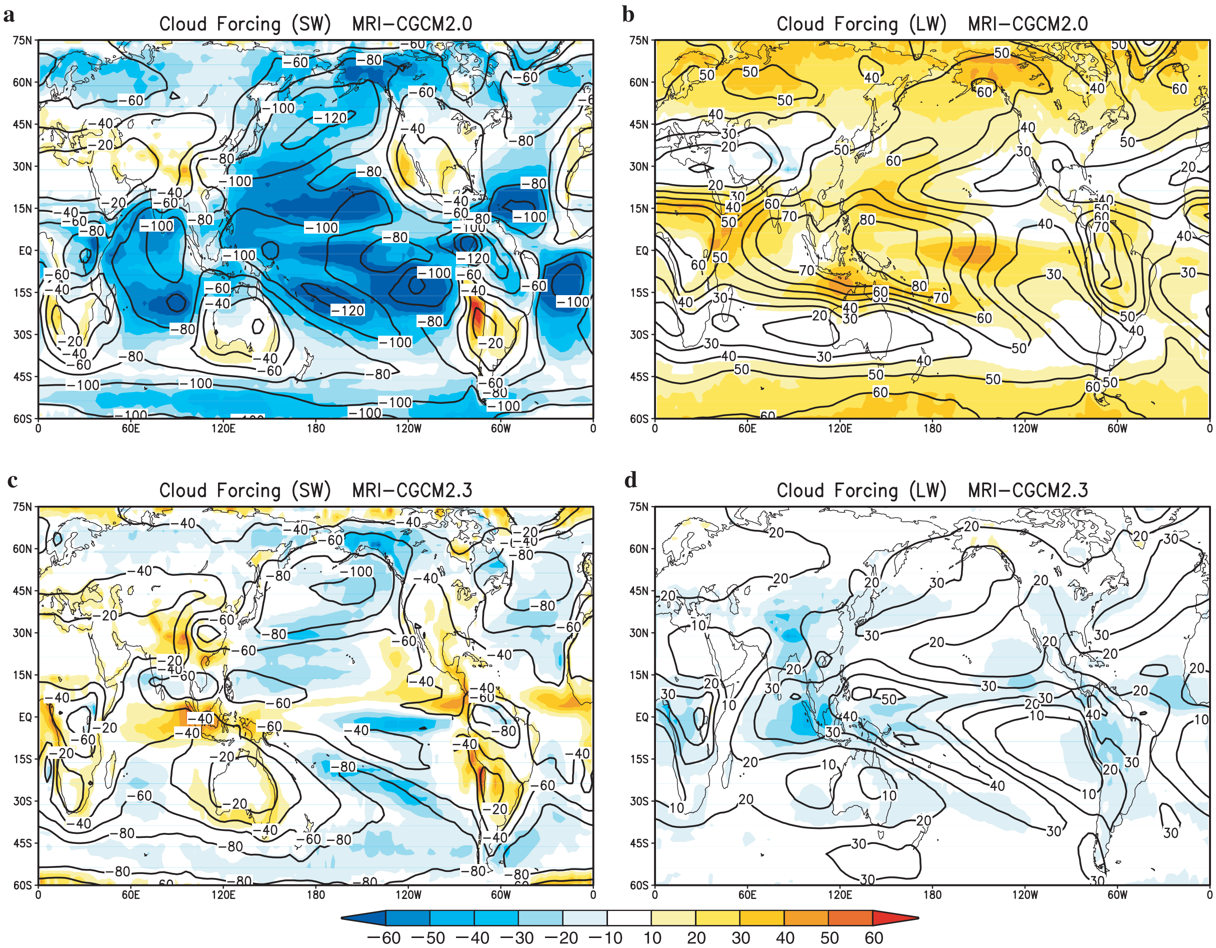


Fig. 12. Geographical distributions of the annual mean cloud forcing (contours) and its deviation (color shadings) from the estimation of the ERBE observation for (a, c) shortwave and (b, d) longwave radiations in the PDcntrl simulations by (a, b) MRI2.0 and (c, d) MRI2.3. Units are Wm^{-2}.

MRI2.3 provided a much better representation of the meridional gradient distribution than MRI2.0, though it was still underestimated in the low to middle latitudes. Overall, the improvements in cloud forcing contributed significantly to the accurate total radiation in MRI2.3 (Fig. 5).

5. Climate sensitivity and cloud response

5.1 Responses in climate sensitivity experiments

The climate responses for the SAT, precipitation, and zonal mean air temperature in the TCR experiments are reviewed in this subsection. These are important fundamentals to be clarified before evaluating climate sensitivity.

a. Surface-air temperature

Temporal variations of the globally averaged SAT in the TCR experiments (1%to4x and 1%to2x), and the equilibrium experiment (2xCO2) with MRI2.3 are illustrated in Fig. 13. The 1%to4x simulation revealed an obvious increase, attaining 15.6°C at the time of CO_2 doubling (year 70), and 17.7°C at the time of CO_2 quadrupling (year 140), with some interannual fluctuations, while the PDcntrl simulation revealed stable behavior that fluctuated around 13.7°C over 500 years. The "cold start" (IPCC 1996) was brief (approximately four years), and it was as sizeable as in MRI2.0's TCR experiment. The temperature increase exhibited very good linearity with respect to time (0.3°C/

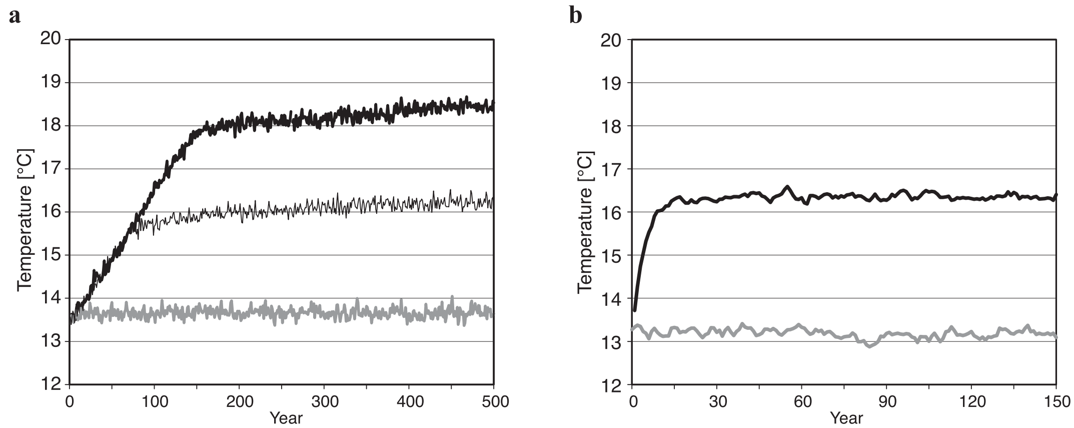


Fig. 13. Temporal evolutions of the globally averaged annual mean SAT for (a) the PDcntrl (thick grey) and TCR (1%to4x: thick solid and 1%to2x: thin solid) experiments with MRI-CGCM2.3 and (b) the equilibrium Slabcntl (grey line) and 2xCO2 (solid line) experiments with MRI-SGCM2.3.

Table 4. Summary of the global climate response for each of the model experiments

Model (Experiment)	MRI-CGCM2.0 (1%to4x)		MRI-CGCM2.3 (1%to4x)		MRI-SGCM2.3 (2xCO2)
CO_2 forcing (average years)	double (61–80)	quadruple (131–150)	double (61–80)	quadruple (131–150)	double (131–150)
Temperature change (K)	1.11	2.19	1.89	3.96	3.17
Precipitation change (%)	1.4	2.4	3.8	8.7	7.2
Effective climate sensitivity (K)	1.39	1.37	2.89	2.82	3.51
Radiative forcing (Wm^{-2})	3.7	7.4	3.7	7.4	3.7
Downward flux at TOA (Wm^{-2})	0.84	1.41	1.04	1.61	0.10
Climate feedback λ ($Wm^{-2}K^{-1}$)	2.69	2.71	1.29	1.32	1.06
Clear-sky feedback λ_{CS} ($Wm^{-2}K^{-1}$)	1.13	1.02	1.18	1.06	0.91
Net cloud feedback λ_{CFNet} ($Wm^{-2}K^{-1}$)	1.56	1.68	0.11	0.25	0.15
SW cloud feedback λ_{CFSW} ($Wm^{-2}K^{-1}$)	0.60	0.67	−0.17	−0.12	0.08
LW cloud feedback λ_{CFLW} ($Wm^{-2}K^{-1}$)	0.96	1.01	0.28	0.37	0.06

decade), and therefore with respect to the logarithm of the CO_2 concentration proportional to its radiative forcing (Wigley 1987). The 20-year averages of the global SAT changes at the times of CO_2 doubling (years 61 to 80), and quadrupling (years 131 to 150) in the 1%to4x simulation were 1.89°C and 3.96°C (Table 4). Results from the similar 1%/yr CO_2 increase experiments (the CMIP2 experiments), using 19 multi-models by IPCC (2001) indicated that the global SAT increase at the time of CO_2 doubling was 1.8°C on average and ranged from 1.1°C to 3.1°C, with a standard deviation 0.4°C. The SAT increase with MRI2.3 was close to this multi-model average value. Additional temperature increases were noted, which were due to an ocean heating committed before stabilizing the CO_2 concentrations at the doubling (for 1%to2x) and quadrupling (for 1%to4x). The equilibrium experiment (2xCO2) indicated a temperature increase of 3.2°C, when sufficient equilibrium was reached. This implies that approximately 40% of equilibrium warming can be additionally expected after stabilization at doubled CO_2 in 1%to2x. The temperature increase at the end of 1%to2x (year 500) was 2.5°C, which is approximately 80% of the equilibrium response. Stouffer and Manabe (1999)

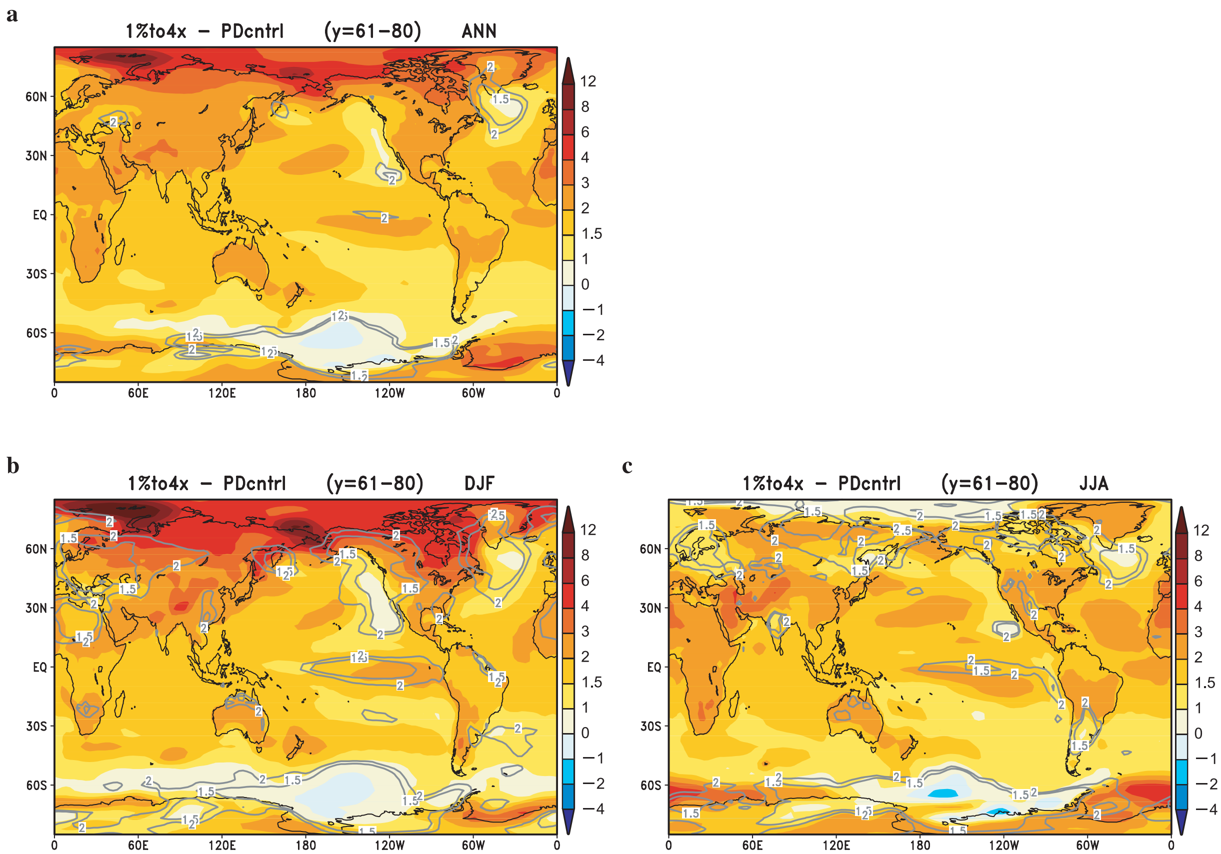


Fig. 14. Geographical distributions of the SAT change (color shading, unit is K) and the change divided by the interannual standard deviation of the PDcntrl (grey contours) in the 1%to4x simulation by MRI2.3 for (a) annual mean, (b) DJF mean, and (c) JJA mean averaged over 20 years centered at the time of CO_2 doubling.

demonstrated that about 35% of the equilibrium response is attained by the time of doubling. Fifteen to twenty centuries may be required for typical coupled models to attain a new equilibrium (IPCC 2001). A considerably larger response fraction appeared to be attained in the same period in our model (~60%), which may have been associated with underestimated ocean-heat uptake. However, there is some uncertainty whether the equilibrium response of a coupled model is consistent with that of a comparable slab mixed-layer ocean model, possibly due, for example, to a difference in the surface temperature in the control climate (Stouffer and Manabe 1999).

Figure 14 depicts the geographical distributions of the SAT changes averaged for 20 years, centered at the time of CO_2 doubling (years 61 to 80) in 1%to4x. Many studies on the transient response to CO_2 increases (e.g., Manabe et al. 1991) have reported that the annual mean warming is relatively substantial over the northern mid high-latitudes, and relatively small over the tropics and the mid-latitude oceans. There is significant warming in the marginal regions of the Arctic Ocean in winter (DJF) in particular, where the sea-ice area remarkably retreats and the thickness is reduced, leading to increased heating by the ocean. The SAT increase is relatively small in the northern North Atlantic and the circum-polar oceans of the southern hemisphere. Deep vertical mixing in these regions is thought to increase the effective heat capacity of the ocean (Manabe et al. 1991). The northern North Pacific exhibits a region with minimum warming in winter, as does the North Atlantic. This is related to the anticyclonic circulation response over the North

Pacific (not shown) as part of the hemispheric annular response that is similar to the Arctic Oscillation (Thompson and Wallace 1998). Warming in the Arctic Ocean is suppressed in summer (Fig. 14c) since the insulating effect of sea ice (which is active in winter, as mentioned above) is absent. This has also been reported in previous studies (e.g., Manabe et al. 1992). The Pacific exhibits a relatively large temperature increase in the central-eastern equatorial region. This feature is an "El-Niño-like" pattern, and is also described in many other studies (e.g., Meehl and Washington 1996; Yamaguchi and Noda 2006).

These response-pattern features have been consistently observed, not only in the multi-model average reported in IPCC (2001), but also in different forcing scenarios (Johns et al. 2003), suggesting the robustness of the SAT response patterns. We examined the geographical pattern correlations of the temperature changes between MRI2.0 and MRI2.3 (Table 5). The correlations were 0.84 (0.91) for the DJF mean, and 0.27 (0.34) for the JJA mean at CO_2 doubling (quadrupling). The correlation for the JJA mean was low, chiefly because there was a substantial difference over land in the northern hemisphere, where MRI2.0 indicated considerably greater warming (with a significant decrease in cloud amount) than MRI2.3 (with little cloud-amount change). The temperature changes in most regions were more than doubled in magnitude compared to the inter-annual standard deviation in the control climate (contours in Fig. 14). The northern North Atlantic, and a portion of the Antarctic Ocean, exhibited a relatively small response ratio. These regions also displayed a relatively small response ratio to the inter-model standard deviation, with the multi-model analysis (IPCC 2001).

Table 5. Global geographical pattern correlations between MRI2.3 and MRI2.0 for the surface air-temperature change ΔTsa and precipitation change ΔPr for the periods of CO_2 doubling (years 61–80) and CO_2 quadrupling (years 131–150)

	DJF mean	JJA mean	Annual mean
ΔTsa at CO_2 doubling	0.84	0.27	0.67
ΔTsa at CO_2 quadrupling	0.91	0.34	0.73
ΔPr at CO_2 doubling	0.12	0.16	0.12
ΔPr at CO_2 quadrupling	0.41	0.13	0.41

b. Precipitation

The rates of precipitation changes at CO_2 doubling in the 1%to4x simulation are depicted in Fig. 15 in geographical distributions for the DJF mean and the JJA mean. Regions with precipitation increase were generally dominant in the tropics in both seasons, while regions with precipitation decrease prevailed in the subtropical latitudes. The precipitation generally increased in the extratropics (particularly in

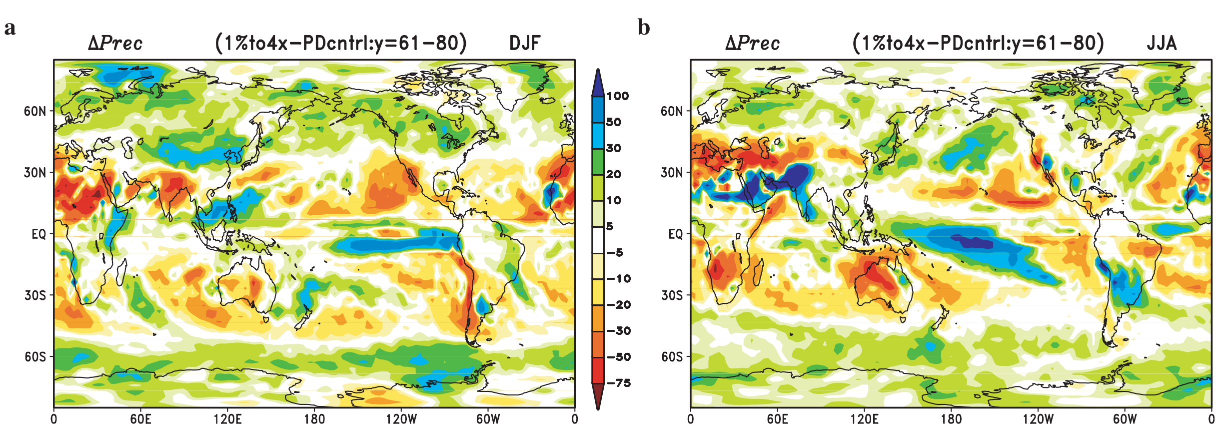


Fig. 15. Geographical distributions of the change rate of precipitation (unit is %) in the 1%to4x experiment with MRI2.3 for (a) DJF mean and (b) JJA mean, averaged over 20 years centered at the time of CO_2 doubling.

the northern winter), associated with enhanced poleward transport of moisture due to the increased water vapor in the warmer atmosphere. We noticed remarkable decreases around the Mediterranean, and in Central Asia in the northern summer (JJA), and a significant percentage increase in the arid region of the southern part of Arabia through Pakistan. These were consistent with the precipitation change in the multi-model ensemble (IPCC 2001), which revealed a general increase in the tropics (particularly the tropical oceans and parts of northern Africa and south Asia), and the mid and high latitudes in response to a CO_2 increase. There was an outstanding increase in the central-eastern tropical Pacific (particularly in JJA). This precipitation anomaly arises from the eastward shift of the active convective rain, associated with the El-Niño-like surface-temperature change (Fig. 14). The feature in the tropical Pacific was also consistent with the results from the multi-models (Yamaguchi and Noda 2006), which also indicate El-Niño-like surface temperature change in most of the models. The El-Niño-like precipitation response patterns differed between DJF and JJA; this was related to the subtle difference of the SST change along the equator, with the maximum shifted eastward in DJF. Decreases of precipitation around the Mediterranean, and the regions of subsidence over the subtropical oceans, were consistently observed in the multi-model ensemble (Figs. 9.11 of IPCC 2001), although it was not explicitly noted in the report. These features indicate that the general pattern of precipitation response in our model was consistent with the average of many other models, suggesting the robustness of the pattern.

The zonally averaged annual mean precipitation changes (Fig. 16) were compared between MRI2.0 and MRI2.3. Both versions showed increases in the tropics and mid and high latitudes, with no change, or only a slight decrease in the subtropics. The increase in the mid latitudes for MRI2.3 was significantly enhanced compared with MRI2.0, and the decrease in the southern subtropics was suppressed. MRI2.3 simulated a greater increase associated with the more prominent El-Niño-like response at the equator, although the difference was smaller than the interannual standard deviation.

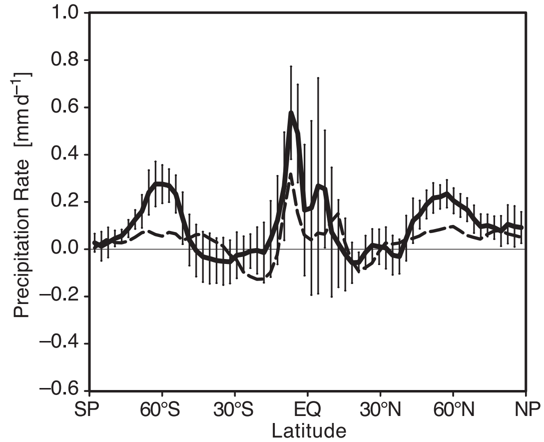


Fig. 16. Meridional distributions of the zonally averaged annual mean precipitation change (unit is mm/day) in the 1%to4x experiments with MRI2.0 (dashed line) and MRI2.3 (thick solid line) averaged over 20 years centered at the time of CO_2 doubling. The vertical bars indicate the ranges of the standard deviation for interannual variation.

The globally averaged annual mean precipitation increased by 3.8% at CO_2 doubling (Table 4), which is significantly enhanced from 1.4% in MRI2.0. The precipitation increase in the equilibrium response to CO_2 doubling (2xCO2) was 7.2%, for a temperature increase of 3.2°C. These responses indicate that the ratio between the changes in temperature and precipitation is consistent with that suggested in climate sensitivity experiments with many models (Figs. 9.18 of IPCC 2001).

c. Zonal mean air temperature

The zonal-mean temperature change in MRI2.3 (Fig. 17a) exhibited a familiar pattern of overall tropospheric warming and stratospheric cooling, with maximum warming at the upper troposphere in the tropics, and the lower troposphere in the northern high-latitudes. This general pattern is consistent with the results of MRI2.0 (not shown) as well as other models in response to CO_2 increase (e.g., Murphy and Mitchell 1995; IPCC 2001). The upper troposphere warming for MRI2.3 in the tropics exceeded 3 K, while that for MRI2.0 was less than 1 K (Fig. 17b). This implies a stronger response of the tropical upper-troposphere

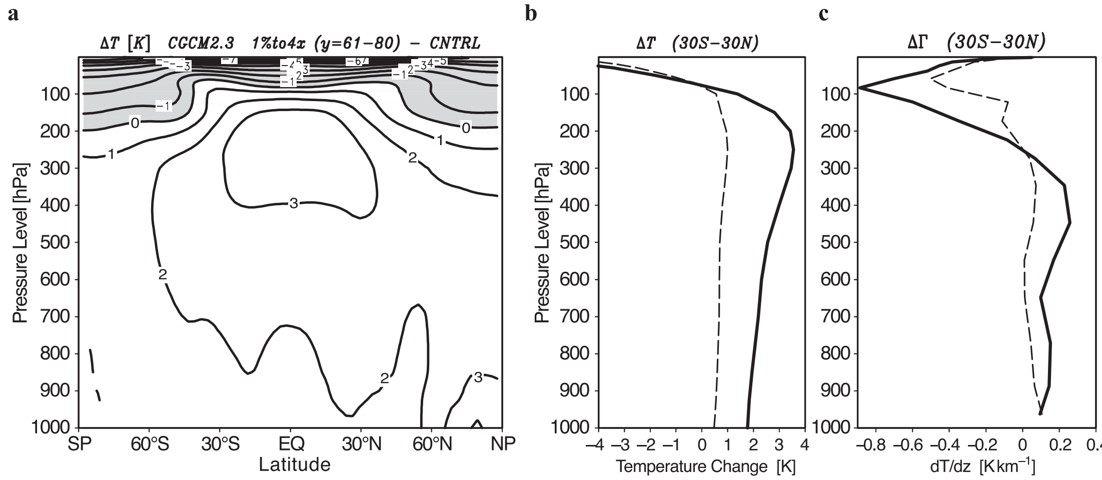


Fig. 17. Zonally averaged annual mean temperature change (unit is K) in the 1%to4x experiment with MRI2.3 averaged over 20 years centered at the time of CO_2 doubling (years 61 to 80) in (a) a meridional-vertical section, and vertical distributions of the changes in (b) temperature (unit is K) and (c) lapse rate (unit is K/km) averaged over the tropical region (30°S–30°N) for MRI2.0 (dashed line) and MRI2.3 (solid line).

temperature in MRI2.3 relative to the global surface-temperature change (tropical upper of ~3 K vs. global surface of 1.9 K) than in MRI2.0 (tropical upper of ~1 K vs. global surface of 1.1 K). MRI2.3 indicated stronger enhancement of the vertical stability in the tropical troposphere at the low to mid levels (Fig. 17c), associated with the greater temperature increase in the upper troposphere than at the surface.

5.2 *Climate sensitivity and feedback*

Identifying a model's climate sensitivity, defined as the equilibrium response of the global mean surface temperature to doubled CO_2 forcing, is very important for evaluating climate changes. It is simple to estimate the climate sensitivity by obtaining an equilibrium response using a full CGCM, though it requires copious computing resources. However, there are alternative methods for the evaluation. One method involves conducting an SGCM equilibrium experiment in which the effects of deep-ocean response are not included and the responses of sea ice differ somewhat from those in full CGCM experiments.

The other method entails calculating the "effective climate sensitivity" (Murphy 1995), using a TCR experiment with CGCM. The effective climate sensitivity is evaluated through a simple heat budget of the climate system, expressed as

$$dH/dt = F - \lambda T$$

where dH/dt is the change rate of heat storage in the system, F is the radiative-forcing change, T is the global mean surface temperature change, and λ represents the net feedback parameter in the system.

Assuming $dH/dt = 0$, the equilibrium climate sensitivity is defined by

$$T_{2x} = F_{2x}/\lambda.$$

Here, F_{2x} is the radiative-forcing change produced by a doubling of the CO_2 concentration. The reciprocal of the effective feedback parameter λ_e is evaluated from a non-equilibrium state in the TCR experiment as

$$1/\lambda_e = T/(F - dH_0/dt) = T/(F - F_0),$$

where dH_0/dt is the ocean-heat uptake and is equal to dH/dt, assuming that heat storage of the atmosphere and land component is negligible and thus F_0 is equal to the net downward-radiation flux at TOA. $1/\lambda_e$ is a measure of feedback magnitude at an arbitrary state in the climate response. Consequently, the effective climate sensitivity is calculated as

$$T_e = F_{2x}/\lambda_e,$$

which is comparable with the equilibrium sensitivity.

The effective climate sensitivities and feedback parameters were calculated for both model versions using this method, and are summarized in Table 4. Fluctuations of ocean-heat uptake lead to a sizeable estimation error for the first several decades, since the calculation includes reciprocal of the small difference between radiative forcing and ocean-heat uptake. Effective climate sensitivity may be time-dependent in a TCR experiment due to inter-hemispheric differences in ocean response (Senior and Mitchell 2000). However, this was not the case for our model since relatively stable values were calculated, with a variation of less than 0.1 K after CO_2 doubling in the 1%to4x simulation. Therefore, we compared climate sensitivities and feedback parameters between MRI2.0 and MRI2.3 primarily for the 20-year average, centered at doubling of the CO_2 in the respective 1%to4x experiment. MRI2.3 exhibited 2.89 K of effective climate sensitivity, which is about twice that of MRI2.0 (1.39 K). The effective climate sensitivity was somewhat less than the equilibrium sensitivity (3.17 K) in the 2xCO2 equilibrium experiment. This sensitivity difference was due to the difference in feedback magnitude, which was most likely associated with a difference in the basic state, particularly for SAT and sea ice. For example, the global mean SAT was 13.1°C in the Slabcntl simulation, which is 0.6°C cooler than that in the PDcntrl simulation, where greater ice-albedo feedback possibly led to greater climate sensitivity (Stouffer and Manabe 1999).

The magnitude of climate sensitivity is significantly affected by feedback from cloud forcing (Cess et al. 1990). We calculated the climate-feedback parameters separately for clear sky (λ_{CS}) and clouds (λ_{CFNet}), to estimate the cloud effects on the climate sensitivity, as expressed below.

$$\lambda_{CS} = (F - F_{CS})/T$$

$$\lambda_{CFNet} = -CF_{Net}/T$$

Here, CF_{Net} is the net cloud-forcing change defined by $CF_{Net} = F_0 - F_{CS}$, where F_{CS} is the flux change in clear-sky radiation. Soden et al. (2004) indicated that estimating cloud feedback using cloud forcing requires some caution, since cloud forcing may contain effects from factors other than cloud changes, and that this is more likely for longwave cloud forcing. However, we adopted a broadly used cloud-forcing method here for comparison. Changes in the cloud-feedback parameter are indicated in Table 4 for each model version.

The clear-sky feedback was mainly attributed to the combined effect of negative feedback by an increase in longwave cooling, and positive feedback by an increase in water vapor (i.e. the greenhouse effect); both act against an increase in surface temperature. Cloud feedback consists of a shading effect in shortwave radiation, and the greenhouse effect in longwave radiation with respect to cloud changes in response to surface warming.

The total feedback in MRI2.0 was $2.69\ \mathrm{Wm^{-2}K^{-1}}$, which is about twice that of $1.29\ \mathrm{Wm^{-2}K^{-1}}$ in MRI2.3, implying a stronger negative feedback in MRI2.0. The clear-sky feedback was similar between the two versions (1.13 and $1.15\ \mathrm{Wm^{-2}K^{-1}}$). In contrast, the large net-cloud feedback in MRI2.0 ($1.56\ \mathrm{Wm^{-2}K^{-1}}$) was significantly reduced in MRI2.3 ($0.11\ \mathrm{Wm^{-2}K^{-1}}$). This suggests that cloud feedback is an essential factor in the difference in climate sensitivity between the versions.

The net cloud feedback can be further separated into shortwave and longwave components, as indicated in Table 4. MRI2.0 had a shortwave component (λ_{CFSW}) of $0.60\ \mathrm{Wm^{-2}K^{-1}}$, and a longwave component (λ_{CFLW}) of $0.96\ \mathrm{Wm^{-2}K^{-1}}$, which indicates negative feedback for both shortwave and longwave. In contrast, the cloud feedback in MRI2.3 indicated weak positive feedback for the shortwave component ($\lambda_{CFSW} = -0.17\ \mathrm{Wm^{-2}K^{-1}}$), and moderate negative feedback for the longwave component ($\lambda_{CFLW} = 0.28\ \mathrm{Wm^{-2}K^{-1}}$). MRI2.3 indicated the opposite sign of cloud feedback between shortwave and longwave, as do many other models (e.g., Colman 2003; Williams et al. 2003). However, both shortwave and longwave cloud feedback were negative in MRI2.0, leading to strong negative total feedback and low climate sensitivity.

The shortwave cloud feedback appeared to be weak negative feedback in the equilibrium experiment with the SGCM of MRI2.3, which differed from that in the TCR experiment with

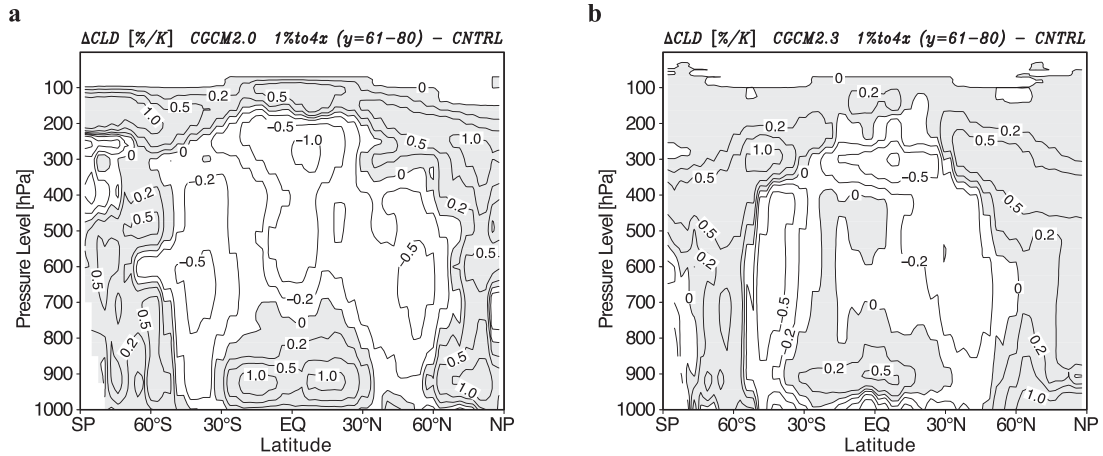


Fig. 18. Meridional-vertical distributions of the zonally averaged annual mean cloud-amount change in the 1%to4x experiments for (a) MRI2.0 and (b) MRI2.3, averaged over 20 years centered at the CO_2 doubling (years 61 to 80). The changes are normalized by the globally averaged SAT change for the corresponding average period. The unit is % K^{-1}.

the CGCM. This was due to a change in clear-sky shortwave radiation, associated with sea-ice changes rather than cloud response. The sea-ice extent was too great in the Slabcntl simulation (particularly in the southern hemisphere), and the decrease in sea-ice area was much greater in the 2xCO2 equilibrium than in the TCR.

5.3 Changes in cloud forcing

The relationship between shortwave and longwave cloud feedback, and the global averages cannot be explained by the cloud response, without distinguishing various cloud categories. Low-level clouds generally exert a substantial shading effect by reflecting more solar radiation if they increase, while their longwave effect is minimal due to the high temperatures of the cloud top. The shortwave effect of high-level clouds is nominal, since there is very little cloud-water content due to the low temperature, while they greatly decrease in OLR if the amount of high clouds increases. It is necessary to examine the cloud response in both the horizontal and vertical distributions to assess each cloud change's contribution to the global cloud feedback.

The meridional-vertical distributions of the cloud response in both versions (Fig. 18) exhibited a general increase in clouds near the tropopause, and a broad decrease in mid-level clouds, except in the Arctic regions. These features of overall cloud response are commonly described in many other model results (e.g., Murphy and Mitchell 1995; Dai et al. 2001). The increase of cloud-top height is primarily attributed to an upward shift of the tropopause by enhanced convection from global warming. It has been suggested that the variation in mid-level clouds is primarily related to changes in mid-troposphere vertical motion (Williams et al. 2003; Bony et al. 2004). Compensating subsidence is enhanced by the enhanced convection in the tropical regions, leading to a decrease in mid-level clouds.

High-level clouds were generally increased in both versions. The increase in the tropics by MRI2.3 above the 200 hPa level appeared to be much less than that by MRI2.0. However, the cloud amount in that region in the control simulation was also much less for MRI2.3, and thus the percentage change (not shown) relative to the present-day value was comparable with MRI2.0.

Different responses of low-level clouds are frequently seen among different models, unlike mid- and high-level clouds. The tropical low-level clouds in MRI2.0 revealed a significant increase around the top of the planetary-boundary layer, while that in MRI2.3 exhibited only a slight increase, and a decrease in the lowest level. We inferred that this difference

Table 6. Area-averaged changes in the cloud forcing (CF) for 1%to4x experiments (20-year averages at CO_2 doubling). Unit is Wm^{-2}.

	MRI-CGCM2.0			MRI-CGCM2.3		
	Global	Tropical (30°S–30°N)	Extra-tropical (poleward of 30°S and 30°N)	Global	Tropical (30°S–30°N)	Extra-tropical (poleward of 30°S and 30°N)
Shortwave CF	−0.83	−1.07	0.23	0.31	1.22	−0.91
Longwave CF	−1.41	−2.18	0.77	−0.54	−1.00	0.47
Net CF	−2.25	−3.25	1.00	−0.23	0.22	−0.44

changed the shortwave cloud-forcing response from a negative value in MRI2.0, toward a positive value in MRI2.3. Williams et al. (2003) suggested that the response of low-level clouds in their CO_2 doubling experiment depended more on the surface-temperature deviation relative to the tropical average, than the change in vertical motions. This implies a dependence of low-level cloud changes on static stability in the lower troposphere. We consider that the greater stability enhancement in MRI2.3 (Fig. 17c) possibly contributed to the decrease of low-level clouds. This will be further discussed later.

The tropical (including subtropical) averages (30°S to 30°N), the extratropical averages (poleward of 30°S and 30°N), and the global averages for shortwave, longwave, and net cloud forcings were compared to examine the relative importance of tropical cloud changes (Table 6). MRI2.3 indicated positive values (0.31 and 1.22 Wm^{-2}) for shortwave cloud-forcing changes for both global and tropical averages, but MRI2.0 yielded negative values (−0.83 and −1.07 Wm^{-2}). The difference in the tropics increased the difference in the global average between the two versions, while the difference in the extratropics partly diminished it. Tropical longwave cloud forcing exhibited a negative change in both MRI2.0 and MRI2.3, though the magnitude of the change was smaller in MRI2.3 than in MRI2.0 by about 1 Wm^{-2}. The difference in the global averages between the versions was similar to that for the tropical averages, implying the relatively small influence of an extratropical longwave cloud-forcing change. The net cloud-forcing change in MRI2.3 yielded a slight positive value (0.22 Wm^{-2}) for the tropical average, and a slight negative value for the global average (−0.23 Wm^{-2}), which resulted from the compensation between the changes of shortwave and longwave forcings with opposite signs. In contrast, the net cloud-forcing change in MRI2.0 had high negative values, which were more evident in the tropics (−3.25 Wm^{-2}). The cloud-forcing change in the tropics is evidently essential to the climate-sensitivity difference between the versions.

Cloud forcing in the tropics critically depends on cloud types, which are generally controlled by atmospheric large-scale circulation. Low-level clouds (stratus and stratocumulus) were dominant in the subsidence regions, such as over the subtropical oceans and the eastern part of the ocean basins. In contrast, convective clouds are produced in the regions of large-scale ascent, with various cloud-top heights from low-level to high-level. Is the cloud change in the climate change also a result of a change in this dynamic control? Is the cloud-forcing response attributable to the change in the large-scale circulation pattern, or the change in the thermodynamic structure of the atmosphere? We conducted analyses similar to Bony et al. (2004), who discretized cloud forcing into circulation regimes, defined by large-scale vertical velocity at 500 hPa, to unravel the dynamic and thermodynamic components of cloud changes.

We defined the statistical weight, P_ω, of each dynamic regime over the tropics (30°S to 30°N), defined as the area covered by regions with a vertical velocity at 500 hPa, or ω_{500}, normalized by the total area of the tropics. The weight, P_ω, is a probability distribution function (PDF) that satisfies

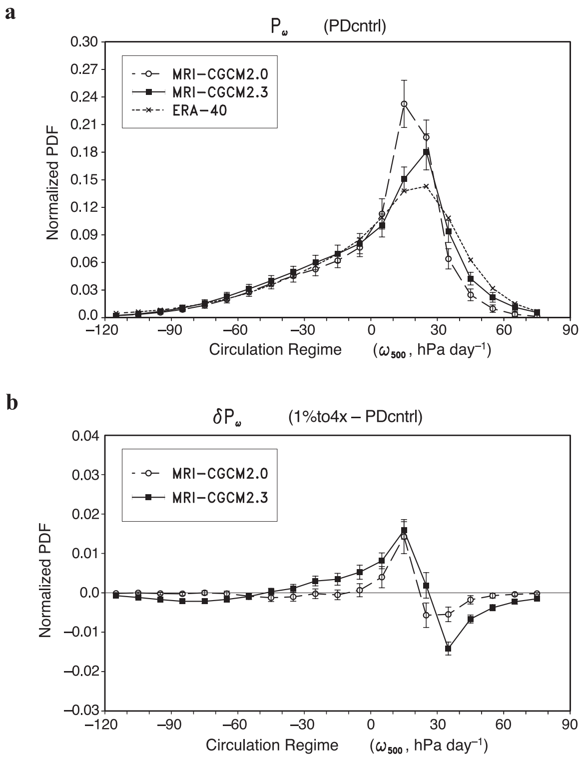


Fig. 19. Normalized PDF of the circulation regime (defined by the vertical velocity at 500 hPa, unit is hPa day^{-1}) in (a) the PDcntrl simulations and (b) its changes (20-year averages at CO_2 doubling) in the 1%to4x experiments for MRI2.0 (dashed line with open circle) and MRI2.3 (solid line with filled square). The dotted line in (a) denotes the PDF for ERA-40.

$$\int P_\omega \, d\omega = 1.$$

Practically, $d\omega$ is calculated for intervals of 10 hPa day^{-1}, based on the monthly mean ω_{500}. Figure 19a compares the PDF P_ω, with respect to the circulation regime ω_{500} for MRI2.0 and MRI2.3, along with that for the atmospheric reanalysis by the European Centre for Medium-range Weather Forecasts (ERA-40; Simmons and Gibson 2000) as a reference. P_ω in the model has a negatively skewed distribution, with a maximum around 10–30 hPa day^{-1} for both versions, which is consistent with the distribution based on ERA-40. This feature suggests that a circulation regime with weak subsidence ($\omega_{500} \sim 10$ to 30 hPa day^{-1}), occupies a large fraction of the entire tropics. A circulation regime with a strong large-scale ascending motion ($\omega_{500} < -40$ hPa day^{-1}), occupies approximately 10% of the tropical area, indicating a strong convective mass flux over regions such as the warm-water pool. MRI2.0 indicated a strong maximum in P_ω at a weaker large-scale subsidence of 10 to 20 hPa day^{-1}, while MRI2.3 indicated a moderate maximum at 20 to 30 hPa day^{-1}, and a greater probability at stronger subsidence ($\omega_{500} > 30$ hPa day^{-1}), which were closer to the P_ω in ERA-40.

We calculated a composite of the cloud forcing C_ω for each different circulation regime defined from ω_{500}. Using this composite, the cloud forcing averaged over the entire tropical region $\bar{C}$ can be expressed as

$$\bar{C} = \int_{-\infty}^{\infty} P_\omega C_\omega \, d\omega,$$

and the area-averaged cloud-forcing change $\overline{\delta C}$ is expressed as

$$\overline{\delta C} = \int_{-\infty}^{\infty} C_\omega \delta P_\omega \, d\omega + \int_{-\infty}^{\infty} \delta C_\omega P_\omega \, d\omega + \int_{-\infty}^{\infty} \delta C_\omega \delta P_\omega \, d\omega.$$

The first rhs term of the above equation is a contribution from changes in the large-scale atmospheric circulation, and is referred to as a dynamic component. The second rhs term is a contribution from changes in the cloud or radiative properties under given dynamic conditions, and is referred to as a thermodynamic component. The third rhs term is the correlation of dynamic and non-dynamic effects in $\overline{\delta C}$.

Figure 20a depicts composites of the cloud forcings, C_ω (for shortwave, longwave, and net), with respect to the circulation regime. The same composites based on the observation (cloud forcings and ω_{500} estimated from the ERBE observation and ERA-40) are also plotted for comparison. The larger longwave cloud forcing corresponds to the stronger large-scale ascending motion, implying the radiative effects of cloud-top variation associated with the convective-activity variation. The longwave cloud forcing in the subsidence regime exhibits weak positive values, and varies little with respect to the subsidence strength. This suggests that low-level clouds are dominant in the subsidence regions and that the cloud-top height depends little on the strength of large-scale circulation. This overall feature was consistent among the two versions of the model and the observation. However, the distribution for MRI2.3 was generally closer to the observation, while that for MRI2.0 suggested an increase in cloud-top height associated with a weakening of subsidence in the region of weak subsidence ($<\sim$30 hPa day^{-1}). Stronger negative shortwave cloud-forcing is generally associated with a stronger large-scale ascending motion. MRI2.3 revealed 20 to 40 Wm^{-2} weaker negative forcings than MRI2.0 and was much closer to the observation at almost all circulation regimes in which the statistical weight was significant ($-40\sim30$ hPa day^{-1}).

We examined the PDF changes, δP_ω, between the averages for the CO_2 doubling in the 1%to4x experiment, and the PDcntrl experiment, to investigate the response of the circulation regime (Fig. 19b). An increase in the weak subsidence ($\omega_{500} < 20$ hPa day^{-1}) and a decrease in the stronger subsidence, were consistently noted in both versions, although MRI2.3 exhibited a slightly larger change than MRI2.0. This indicated a weakening of the large-scale tropical circulation and was also consistent with the results from a uniformly increased SST experiment using other models (Bony et al. 2004). This circulation change resulted in a similar dynamic component of the cloud-forcing change, $C_\omega \delta P_\omega$, for both versions (Fig. 20b), indicating a net forcing change with negative values for $\omega_{500} < 20$ hPa day^{-1}, and positive values for $\omega_{500} > 20$ hPa day^{-1} in the subsidence regimes, which are both dominated by shortwave contributions. We noted a positive contribution to the net cloud-forcing change in MRI2.3 associated with a decrease in the strong ascent in MRI2.3 in the large-scale ascending-motion regimes, when $\omega_{500} < -50$ hPa day^{-1} (strong convective regions) (Fig. 19b), though it was not noted in MRI2.0.

We next calculated δC_ω (not shown) to examine how cloud forcings change in the same circulation regime in response to a climate change. There were significant differences in the cloud-forcing changes between MRI2.0 and MRI2.3 in the strong ascending regimes as well as in the subsidence regimes. However, substantial contributions to the climate sensitivity must be evaluated by $\delta C_\omega P_\omega$ (Fig. 20c), since intermediate regimes provide a significant contribution due to the large statistical weight. Consequently, there is a substantial difference in the weak subsidence regimes, in which the cloud forcings in MRI2.3 indicated a positive change for shortwave (weakened negative shortwave cloud forcing), but almost no change for longwave, resulting in a positive net change. In contrast, the cloud forcings in the weak subsidence regimes in MRI2.0 indicated significant negative changes for both shortwave and longwave, resulting in a strong negative net forcing change. The differences between the versions were minimal in the ascending motion regimes. We also examined the correlation terms $\delta C_\omega \delta P_\omega$ (Fig. 20d) and found them to provide generally small contributions, with a slight damping effect on the dynamic component (Fig. 20b) in the subsidence regimes for both versions.

The above results indicate that cloud-forcing changes are essential in the circulation regimes of tropical weak subsidence in response to climate changes. The contribution of the dynamic component in these circulation regimes produced a similar response between the versions, which suggests that the thermodynamic effects are responsible for the difference in low-level cloud changes under a similar dynamic effect (increase in weaker subsidence). Bony et al. (2004) demonstrated that the thermodynamic component explains most of the tropical averaged cloud-forcing response to an imposed perturbation, and that the thermodynamic component strongly depends on the behavior of the low-level clouds that occur in regions of moderate subsidence. Our results are consistent with their argument.

6. Summary and discussion

MRI-CGCM2 was developed from an early version (MRI-CGCM2.0; Yukimoto et al. 2001) and was released as the new version of the model (MRI-CGCM2.3). The modifications primarily address the cloud scheme; the budgets at the TOA for each shortwave, longwave, and net radiation in the present-day control climate simulation exhibit good agreement with the observations. The simulated cloud amount was remarkably reduced in the low-level clouds over the tropical oceans in association with this

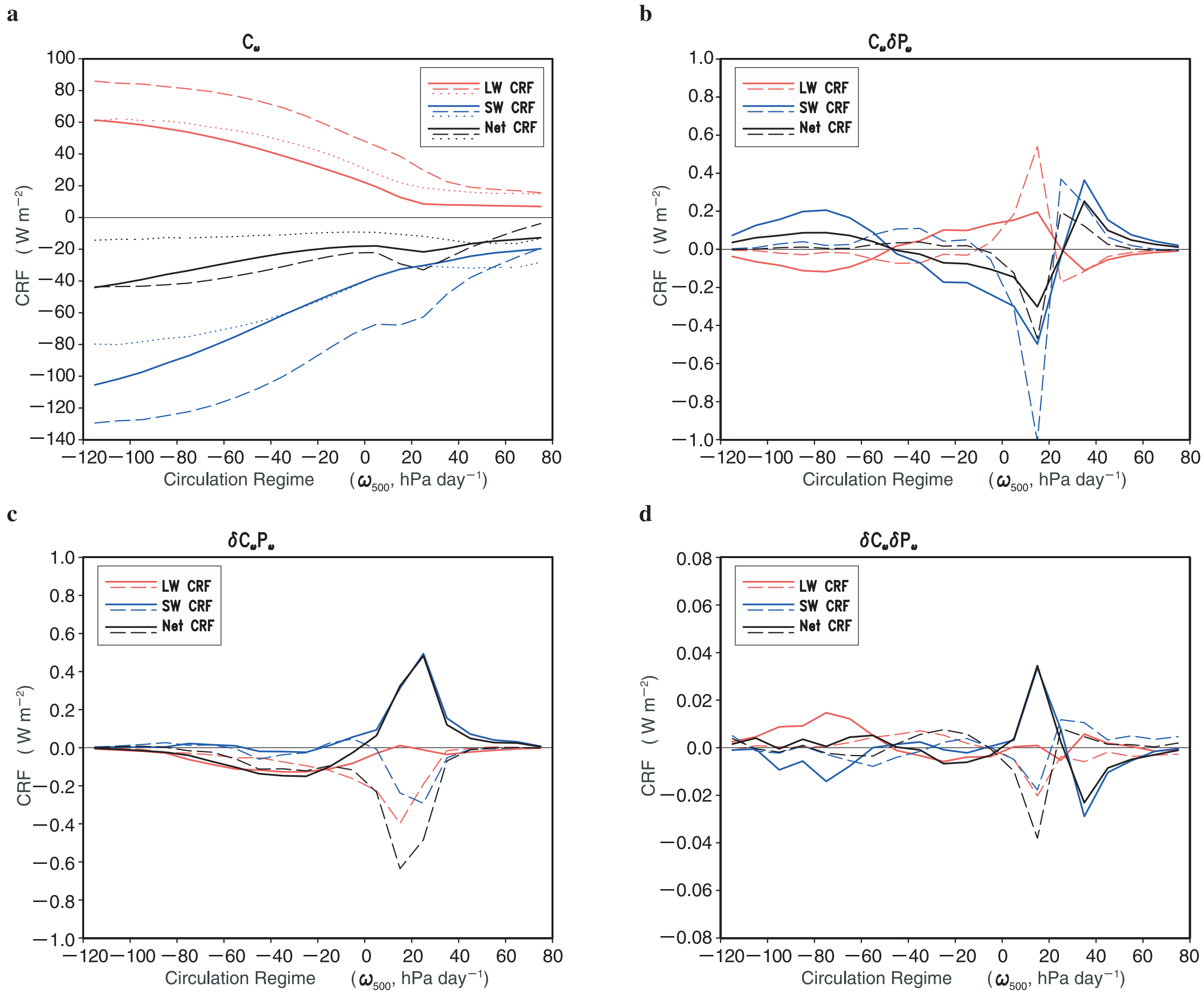


Fig. 20. Composites for (a) C_ω, (b) $C\omega\delta P_\omega$, (c) $\delta C_\omega P_\omega$ and (d) $\delta C_\omega\delta P_\omega$ with respect to the circulation regime for shortwave (SW, blue), longwave (LW, red), and net (Net, black) CRF (unit is Wm^{-2}) in the experiments with MRI2.0 (dashed lines) and MRI2.3 (solid lines). The observational estimations for C_ω are also plotted (a, dotted lines). See the text for definitions of C_ω, $C_\omega\delta P_\omega$, $\delta C_\omega P_\omega$, and $\delta C_\omega\delta P_\omega$.

change, which contributes to decreasing the overestimated reflected shortwave radiation. The cloud amount was also reduced globally around the tropopause, which lowered the cloud top and increased the OLR.

An accurate radiation budget in both the global average and meridional distribution, enables us to obtain realistic values for the implied ocean meridional heat transport at each latitude, even if the flux adjustment (for heat) is not included. We tested the model without employing any flux adjustment for heat (not shown in the present paper), and found that the simulated SST agreed reasonably with the observation, and had sufficiently small climate drift.

The improved cloud and radiation budget also contributed to better representation of the SAT. There was a substantial warm bias over land in summer in the previous version, which was mostly reduced in the current version. The precipitation distribution also approached the observed climate, probably in association with the more realistic radiation. Heating due to condensation precipitation was primarily balanced, with the column radiative cooling of a large scale in the tropics, and thus we may be able to obtain a realistic precipitation distribution if we impose an idealized radiative-cooling distribution. However, it is difficult to control the idealized balance, since there is a strong interaction between the condensation and large-scale circulation. Nonetheless, creating a realistic balance between clouds and radiation is important for improving the precipitation distribution as well as for modifying the convective scheme in the model. One marked improvement in the precipitation distribution is the alleviation of the double-ITCZ problem in the model. Philander et al. (1996) indicated that the equatorial asymmetry over the tropical Pacific is related to an air-sea interaction in the eastern Pacific. They also suggested that the lack of the important tropical stratus process that is involved in the interaction, may be responsible for the overly strong double ITCZ simulated in the model. The associated air-sea interaction would most likely become realistic given the proper distribution of low-level clouds in the eastern oceans in our new version, which may contribute to reducing the double-ITCZ tendency.

We analyzed the climate sensitivity by examining changes in the TCR experiments with increasing the CO_2 at a rate of 1%/year. The effective climate sensitivity estimated from the TCR experiment was 2.9 K for MRI2.3, which was an increase of approximately twice that for MRI2.0. The response of the global mean precipitation indicated a 3.8% increase around the CO_2 doubling, which was also significantly greater than in the previous version with its 1.4% increase.

Warming over the continents in summer is suppressed in the new version, despite the increased warming of the annual mean global temperature. This is probably attributable to an increase in the cloud amount, with negative shortwave cloud forcing in the region in summer, which leads to an increase of precipitation, increase of soil moisture, and enhanced evaporation, which acts as a positive feedback loop.

The difference in cloud-forcing response is significant in the climate-sensitivity difference between the model versions. Although the climate feedback from clear-sky radiation was almost the same value ($\sim 1\ \mathrm{Wm^{-2}K^{-1}}$) between the two versions, the net cloud feedback was much smaller in MRI2.3 ($0.1 \sim 0.3\ \mathrm{Wm^{-2}K^{-1}}$) than in MRI2.0 ($1.6 \sim 1.7\ \mathrm{Wm^{-2}K^{-1}}$), accounting for most of the difference in the total climate feedback ($1.3\ \mathrm{Wm^{-2}K^{-1}}$ for MRI2.3 vs. $2.7\ \mathrm{Wm^{-2}K^{-1}}$ for MRI2.0).

This difference in the cloud feedback can be attributed to the difference in cloud-forcing response, which is dominated by the tropical clouds for both shortwave and longwave cloud forcings. We investigated the origin of the difference in the cloud-forcing response in terms of whether it is mainly controlled by a difference in large-scale atmospheric circulation changes or by a difference in the thermodynamic effects. The cloud forcings were discretized based on the large-scale circulation regime, represented by the vertical velocity at 500 hPa. The model simulated the proper PDF of the cloud forcing, which revealed that the regions with weak subsidence dominate the entire tropics, in accordance with the observational PDF. We examined changes in the tropical large-scale circulation, using changes in the PDF of the vertical velocity. The tropical large-scale circulation was found to be generally weakened in response to the CO_2 increase; this was consistent for both versions. The low-level cloud amount decreased and the negative shortwave cloud forcing decreased in MRI2.3 under a similar circulation change with weakened subsidence. In contrast, an increase in the low-level cloud amount in addition to an increase in the cloud-top height in MRI2.0, led to enhanced negative shortwave cloud forcing and decreased positive longwave cloud forcing. The tropical cloud response appears to be controlled more by thermodynamic effects, than by dynamic effects. The difference in stabilization of the tropical lower troposphere (Fig. 17c), is probably one of the most important thermodynamic effects of the cloud response. Furthermore, the difference in thermodynamic effects may be related to the climatological dependence of low-level clouds on the subsidence strength. Both the shortwave and longwave cloud forcings were independent of the subsidence strength in the control climate of the MRI2.3 and the ERBE observations, unlike in MRI2.0 in which the shortwave and longwave cloud forcings tended to be enhanced as the subsidence weakened (Fig. 20a). This may explain why MRI2.0

exhibited negative cloud forcings for both shortwave and longwave under weakened subsidence in an increased atmospheric CO_2 concentration.

This study demonstrated that improving the simulated cloud forcing in the control climate increases the climate sensitivity (and hence decreases the negative cloud feedback) of the model. However, we cannot conclude that this accurately reflects the reality. Cloud changes in response to increased greenhouse gases would be affected by changes in many processes, including cloud microphysics, which were not formulated in the current model parameterization. More detailed analyses are required in future studies. We focused on the cloud response associated with climate sensitivity in the present study, and thus we investigated it primarily based on the global (or tropical) averages and the annual means, and discovered that a change in the tropical average is essential. However, we must pay more attention to the changes in the extratropics as regional climate changes or climate-change patterns, for which examining the seasonal base is more important. For example, the difference in cloud response over the summer continents between the model versions should be investigated in more detail.

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References

Adler, R.F., G.J. Huffman, A. Chang, R. Ferraro, P. Xie, J. Janowiak, B. Rudolf, U. Schneider, S. Curtis, D. Bolvin, A. Gruber, J. Susskind, P. Arkin, and E. Nelkin, 2003: The Version-2 Global Precipitation Climatology Project (GPCP) Monthly Precipitation Analysis (1979-Present). *J. Hydrometeorol.*, **4**, 1147–1167.

Arakawa, A. and W.H. Schubert, 1974: Interaction of a cumulus cloud ensemble with the large scale environment, Part I. *J. Atmos. Sci.*, **31**, 674–701.

Barkstrom, B., E. Harrison, G. Smith, R. Green, J. Kibler, R. Cess, and the ERBE Science Team, 1989: Earth Radiation Budget Experiment (ERBE) archival and April 1985 results. *Bull. Amer. Meteor. Soc.*, **70**, 1254–1262.

Bony, S., J.-L. Dufresne, H. Le Treut, J.-J. Morcrette, and C. Senior, 2004: On dynamic and thermodynamic components of cloud changes. *Clim. Dyn.*, **22**, 71–86.

Bourke, R.H. and R.P. Garrett, 1987: Sea ice thickness distribution in the Arctic Ocean. *Cold Reg. Sci. Technol.*, **13**, 259–280.

Briegleb, B.P., 1992: Delta-Eddington approximation for solar radiation in the NCAR community climate model. *J. Geophys. Res.*, **97**, 7603–7612.

———, P. Minnis, V. Ramanathan, and E. Harrison, 1986: Comparison of regional clear-sky albedos inferred from satellite observations and model computations. *J. Appl. Meteor.*, **25**, 214–226.

Cess, R.D., G.L. Potter, J.P. Blanchet, G.J. Boer, A.D. Del Genio, M. Déqué, V. Dymnikov, V. Galin, W.L. Gates, S.J. Ghan, J.T. Kiehl, A.A. Lacis, H. Le Treut, B.J. McAvaney, V.P. Meleshko, J.F.B. Mitchell, J.-J. Morcrette, D.A. Randall, L. Rikus, E. Roeckner, J.F. Royer, U. Schlese, D.A. Sheinin, A. Slingo, A.P. Sokolov, K.E. Taylor, W.M. Washington, R.T. Wetherald, and I. Yagai, 1990: Intercomparison and interpretation of climate feedback processes in 19 atmospheric general circulation models. *J. Geophys. Res.*, **95**(D10), 16601–16615, 10.1029/90JD01219.

Colman, R., 2003: A comparison of climate feedbacks in general circulation models. *Clim. Dyn.*, **20**, 865–873.

Dai, A., T.M.L. Wigley, A. Boville, J.T. Kiehl, and L.E. Buja, 2001: Climates of the twentieth and twenty-first Centuries simulated by the NCAR Climate System Model. *J. Climate*, **14**, 485–519.

Ganachaud, A. and C. Wunsch, 2000: Improved estimates of global ocean circulation, heat transport and mixing from hydrographic data. *Nature*, **408**, 453–457.

Gent, P.R. and J.C. McWilliams, 1990: Isopycnal mixing in ocean circulation models. *J. Phys. Oceanogr.*, **20**, 150–155.

Harrison, E.F., P. Minnis, B.R. Barkstrom, V. Ramanathan, R.D. Cess, and G.G. Gibson, 1990: Seasonal variation of cloud radiative forcing derived from the Earth radiation budget experiment. *J. Geophys. Res.*, **95**(D11), 18,687–18,703.

IPCC, 1996: *Climate Change 1995: The Science of Climate Change, Contribution of Working Group I to the Second Assessment Report of the Intergovernmental Panel on Climate Change*, (Eds. Houghton, J.T., L.G. Meira Filho, B.A. Callander, N. Harris, A. Kattenberg, and K. Maskell), Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 572pp.

———, 2001: *Climate Change 2001: The Scientific Basis, Intergovernmental Panel on Climate Change, Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change*, (Eds. Houghton, J.T., Y. Ding, D.J. Griggs, M. Noguer, P.J. van der Linden, X. Dai, K. Maskell, and C.A. Johnson), Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 881pp.

Iwasaki, T., S. Yamada, and K. Tada, 1989: A parameterization scheme of orographic gravity wave drag with the different vertical partitioning, part 1: Impact on medium range forecasts. *J. Meteor. Soc. Japan*, **67**, 11–41.

Johns, T.C., J.M. Gregory, W.J. Ingram, C.E. Johnson, A. Jones, J.A. Lowe, J.F.B. Mitchell, D.L. Roberts, D.M.H. Sexton, D.S. Stevenson, S.F.B. Tett, and M.J. Woodage, 2003: Anthropogenic climate change for 1860 to 2100 simulated with the HadCM3 model under updated emissions scenarios. *Clim. Dyn.*, **20**, 583–612.

Jones, P.D., M. New, D.E. Parker, S. Martin, and I.G. Rigor, 1999: Surface air temperature and its variations over the last 150 years. *Reviews of Geophysics*, **37**, 173–199.

Kang, I.-S., K. Jin, B. Wang, K.-M. Lau, J. Shukla, V. Krishnamurthy, S. Schubert, D. Wailser, W. Stern, A. Kitoh, G.A. Meehl, M. Kanamitsu, V. Galin, V. Satyan, C.-K. Park, and Y. Liu, 2002: Intercomparison of the climatological variations of Asian summer monsoon precipitation simulated by 10 GCMs. *Clim. Dyn.*, **19**, 383–395.

Kiehl, J.T. and K.E. Trenberth, 1997: Earth's Annual Global Mean Energy Budget. *Bull. Amer. Meteorol. Soc.*, **78**, 197–208.

Langner, J. and H. Rodhe, 1991: A global three-dimensional model of the tropospheric sulfur cycle. *J. Atmos. Chem.*, **13**, 225–263.

Levitus, S. and T.P. Boyer, 1994: World Ocean Atlas, Volume 4: Temperature, NOAA Atlas NESDIS 4, 129 pp.

———, R. Burgett, and T.P. Boyer, 1994: *World Ocean Atlas, Volume 3: Salinity*, NOAA Atlas NESDIS 3, 111 pp.

Manabe, S., R.J. Stouffer, M.J. Spelman, and K. Bryan, 1991: Transient responses of a coupled ocean-atmosphere model to gradual changes of atmospheric CO_2. Part I: Annual mean response. *J. Climate*, **4**, 785–818.

———, M.J. Spelman, and R.J. Stouffer, 1992: Transient responses of a coupled ocean-atmosphere model to gradual changes of atmospheric CO_2. Part II: Seasonal response. *J. Climate*, **5**, 106–126.

Meehl, G.A. and W.M. Washington, 1996: El Niño-like climate change in a model with increased atmospheric CO_2-concentrations. *Nature*, **382**, 56–60.

Mellor, G.L. and P.A. Durbin, 1975: The structure and dynamics of the ocean surface mixed layer. *J. Phys. Oceanogr.*, **5**, 718–728.

——— and L. Kantha, 1989: An ice-ocean coupled model. *J. Geophys. Res.*, **94**, 10937–10954.

——— and T. Yamada, 1974: A hierarchy of turbulence closure models for planetary boundary layers. *J. Atmos. Sci.*, **31**, 1791–1806.

——— and ———, 1982: Development of a turbulence closure model for geophysical fluid problems. *Rev. Geophys. Space Phys.*, **20**, 851–875.

Mizuta, R., K. Oouchi, H. Yoshimura, A. Noda, K. Katayama, S. Yukimoto, M. Hosaka, S. Kusunoki, H. Kawai, and M. Nakagawa, 2006: 20-km-mesh global climate simulations using JMA-GSM model—mean climate states—. *J. Meteor. Soc. Japan*, **84**, 165–185.

Murphy, J.M., 1995: Transient response of the Hadley Centre coupled ocean-atmosphere model to increasing carbon dioxide. Part III: Analysis of global-mean response using simple models—. *J. Climate*, **8**, 496–514.

——— and J.F.B. Mitchell, 1995: Transient response of the Hadley Centre coupled ocean-atmosphere model to increasing carbon dioxide, Part II: Spatial and temporal structure of response. *J. Climate*, **8**, 57–80.

Oouchi, K., J. Yoshimura, H. Yoshimura, R. Mizuta, S. Kusunoki, and A. Noda, 2006: Tropical cyclone climatology in a global-warming climate as simulated in a 20 km-mesh global atmospheric model: frequency and wind intensity analyses. *J. Meteor. Soc. Japan*, **84**, 259–276.

Philander, S.G.H., D. Gu, G. Lambert, T. Li, D. Halpern, N.-C. Lau, and R.C. Pacanowski, 1996: Why the ITCZ is mostly north of the equator. *J. Climate*, **9**, 2958–2972.

Pierrehumbert, R.T., 1995: Thermostats, radiator fins, the local runaway greenhouse. *J. Atmos. Sci.*, **52**, 1784–1806.

Ramanathan, V. and W. Collins, 1991: Thermodynamic regulation of ocean warming by cirrus clouds deduced from observations of the 1987 El Niño. *Nature*, **351**, 27–32.

Randall, D. and D.-M. Pan, 1993: Implementation of the Arakawa-Schubert cumulus parameterization with a prognostic closure, Meteorological Monograph/The representation of cumulus convection in numerical models, **46**, 145–150.

Rajendran, K., A. Kitoh, and S. Yukimoto, 2004: South and East Asian monsoon climate and variation in the MRI coupled model (MRI-CGCM2). *J. Climate*, **17**, 763–782.

Rayner, N.A., D.E. Parker, E.B. Horton, C.K. Folland, L.V. Alexander, D.P. Rowell, E.C. Kent, and A. Kaplan, 2003: Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *J. Geophys. Res.*, **108**(D14), 4407, doi:10.1029/2002JD002670.

Sato, N., P.J. Sellers, D.A. Randall, E.K. Schneider, J. Shukla, J.L. Kinter, Y.-Y. Hou, and E. Albertazzi, 1989: Effects of implementing the simple biosphere model in a general circulation model. *J. Atmos. Sci.*, **46**, 2757–2782.

Sellers, P.J., Y. Mintz, Y.C. Sud, and A. Dalcher, 1986: A simple biosphere model (SiB) for use within general circulation models. *J. Atmos. Sci.*, **43**, 505–531.

Senior, C.A. and J.F.B. Mitchell, 2000: The time-dependence of climate sensitivity. *Geophys. Res. Lett.*, **27**(17), 2685–2688.

Shettle, E.P. and R.W. Fenn, 1979: *Models for the aerosols of the lower atmosphere and the effects of humidity variation on their optical properties*, AFGL-TR-79-0214, Air Force Geophysical Laboratory, 94pp.

Shibata, K. and T. Aoki, 1989: An infrared radiative scheme for the numerical models of weather and climate. *J. Geophys. Res.*, **94**, 14923–14943.

——— and A. Uchiyama, 1992: Accuracy of the delta-four-stream approximation in inhomogeneous scattering atmospheres. *J. Meteor. Soc. Japan*, **70**, 1097–1109.

———, H. Yoshimura, M. Ohizumi, M. Hosaka, and M. Sugi, 1999: A simulation of troposphere, stratosphere and mesosphere with an MRI/JMA98 GCM. *Pap. Meteor. Geophys.*, **50**, 15–53.

Slingo, A., l989: A GCM parameterization for the shortwave radiative properties of water clouds. *J. Atmos. Sci.*, **46**, 1419–1427.

Simmons, A.J. and J.K. Gibson, 2000: The ERA-40 Project plan, ERA-40 Proj. Rep. Ser. 1, 63 pp., Eur. Cent. for Medium-Range Weather Forecasts, Reading, UK.

Soden, B.J., A.J. Broccoli, and R.S. Hemler, 2004: On the use of cloud forcing to estimate cloud feedback. *J. Climate*, **17**, 3661–3665.

Stephens, G.L., 1978: Radiation profiles in extended water cloud. II: parameterization schemes. *J. Atmos. Sci.*, **35**, 2123–2132.

———, 2005: Cloud feedbacks in the climate system: A critical review. *J. Climate*, **18**, 237–273.

———, S.-C. Tsay, P.W. Stackhouse, Jr., and P.J. Flatau, 1990: The relevance of the microphysical and radiative properties of cirrus clouds to climate and climatic feedback. *J. Atmos. Sci.*, **47**, 1742–1753.

Stouffer, R.J. and S. Manabe, 1999: Response of a coupled ocean-atmosphere model to increasing atmospheric carbon dioxide: Sensitivity to the rate of increase. *J. Climate*, **12**, 2224–2237.

Tegen, I., P. Hollrig, M. Chin, I. Fung, D. Jacob, and J. Penner, 1997: Contribution of different aerosol species to the global aerosol extinction optical thickness: Estimates from model results. *J. Geophys. Res.* **102**, 23895–23915.

Thompson, D.W.J. and J.M. Wallace, 1998: The Arctic Oscillation signature in the winter time geopotential height and temperature fields. *Geophys. Res. Lett.*, **25**, 1297–1300.

Tsujino, H., H. Hasumi, and N. Suginohara, 2000: Deep pacific circulation controlled by vertical diffusivity at the lower thermocline depth. *J. Phys. Oceanogr.*, **30**, 2853–2865.

Wigley, T.M.L., 1987: Relative contributions of different trace gases to the greenhouse effect. *Climate Monitor*, **16**, 14–29.

Williams, K.D., M.A. Ringer, and C.A. Senior, 2003: Evaluating the cloud response to climate change and current climate variability. *Clim. Dyn.*, **20**, 705–721.

Yamaguchi, K. and A. Noda, 2006: Global warming patterns over the North Pacific: ENSO versus AO. *J. Meteor. Soc. Japan*, **84**, 221–241.

Yukimoto, S. and Y. Kitamura, 2003: An investigation of the irregularity of El Niño in a coupled GCM. *J. Meteor. Soc. Japan*, **81**, 599–622.

———, A. Noda, A. Kitoh, M. Sugi, Y. Kitamura, M. Hosaka, K. Shibata, S. Maeda, and T. Uchiyama, 2001: The New Meteorological Research Institute Coupled GCM (MRI-CGCM2)—Model Climate and Variability—. *Pap. Meteor. Geophys.*, **51**, 47–88.

———, ———, T. Uchiyama, S. Kusunoki, and A. Kitoh, 2005: Climate changes of the twentieth through twenty-first centuries simulated by the MRI-CGCM2.3. *Pap. Meteor. Geophys*. (in press).